Inventor 2040 SMARS Build Guide

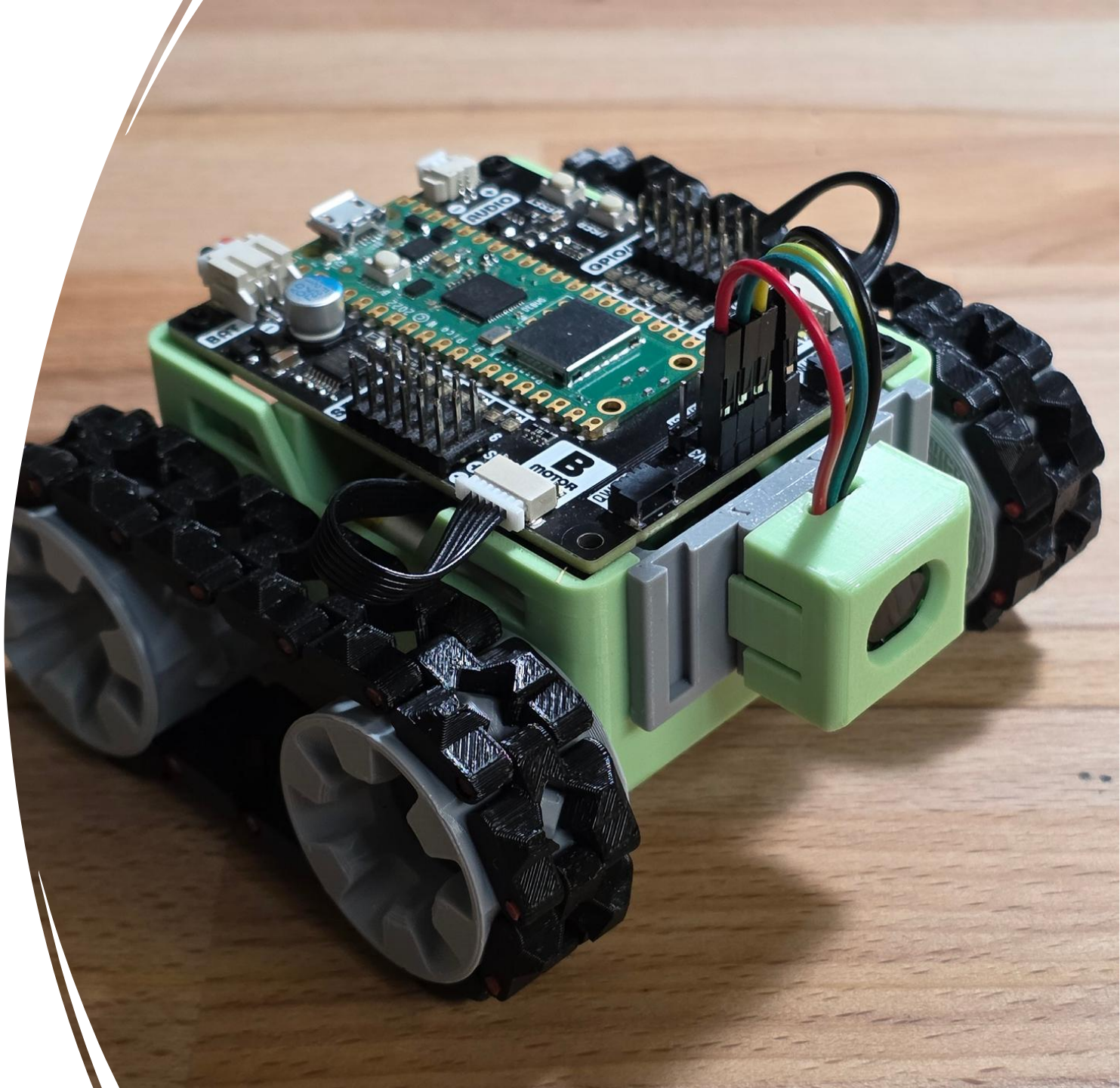


Table of Contents

Build Guide:

1. Installing Charging PCBA and Battery
2. Install Motors
3. Install Battery Plate
4. Install Gearhead Motor Encoder Cables
5. Install Front Mounting Bracket
6. Assembling Speaker to Inventor 2040 PCBA
7. Installing Inventor 2040 PCBA
8. Install Slave Wheels
9. Assembling Tank Tracks
10. Installing Drive Wheels and Tank Tracks
11. Assembling VX53L1x Time of Flight (TOF) Sensor
12. Installing the VX53L1x TOF Sensor Assembly
13. Testing and Code Installation: Getting your SMARS to Come Alive

Appendix A: Component Information
Appendix B: Various Part Drawings

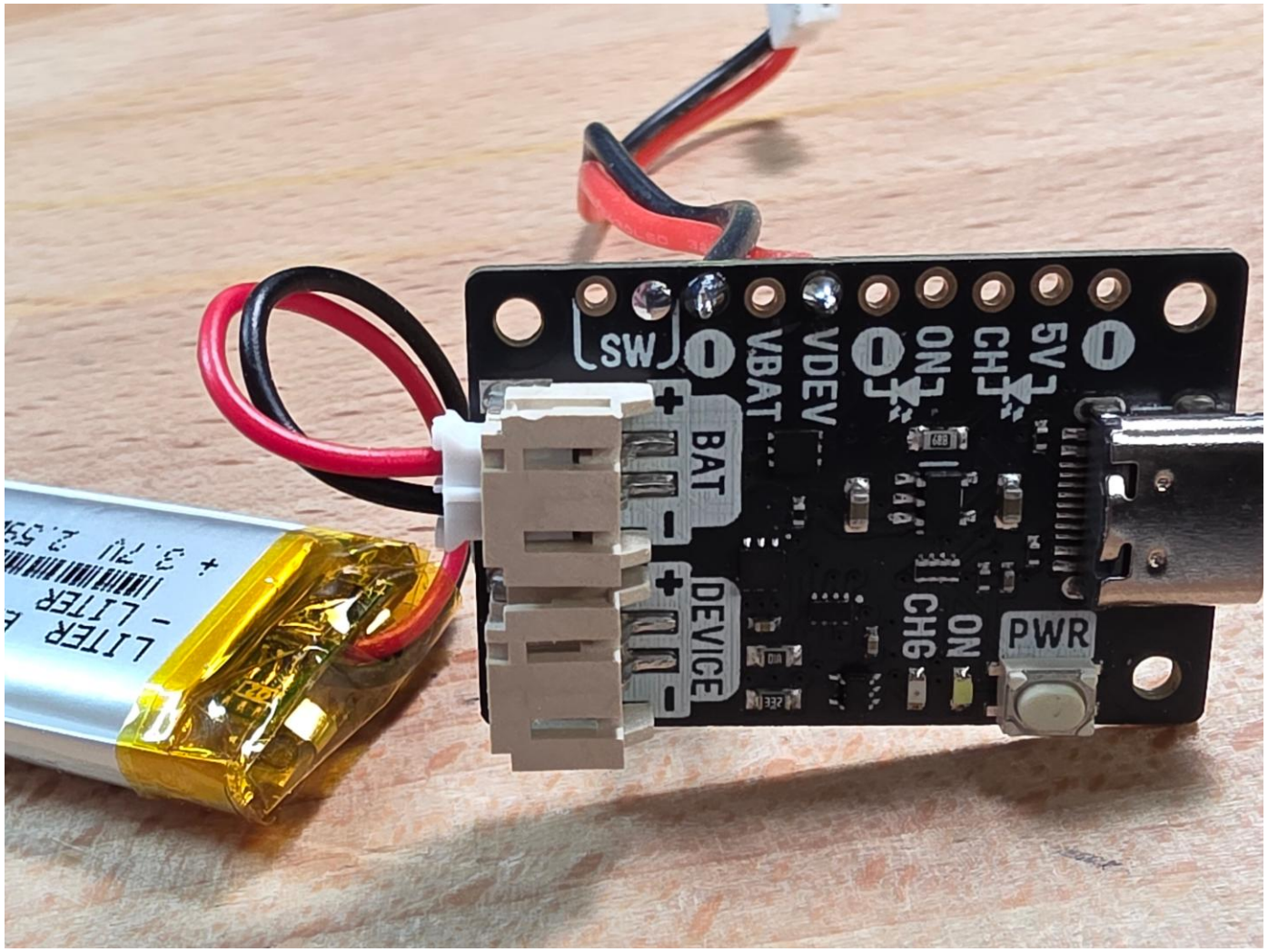
Installing Charging PCBA
and Battery

Plug the battery connector into the LiPo Amigo Pro Charging PCBA

Note: Polarity and Connector Orientation.

Test by pushing the PWR button. If correct the “ON” LED will light up white when on. Turn off before next steps.

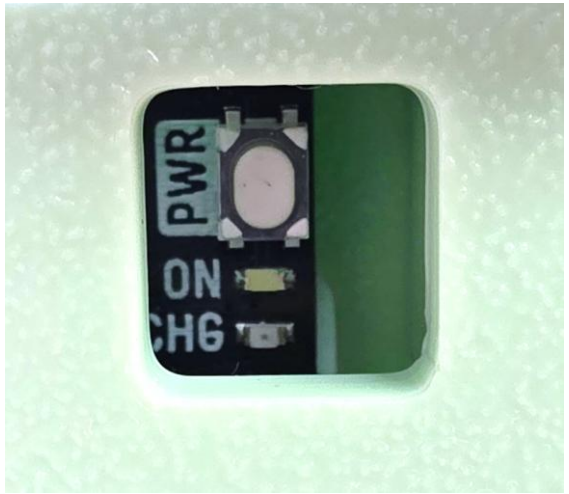
Next plug into a USB C power source. The “CHG” LED should Illuminate red until battery is charged. Battery supplied was charged so the LED may not stay on very long.

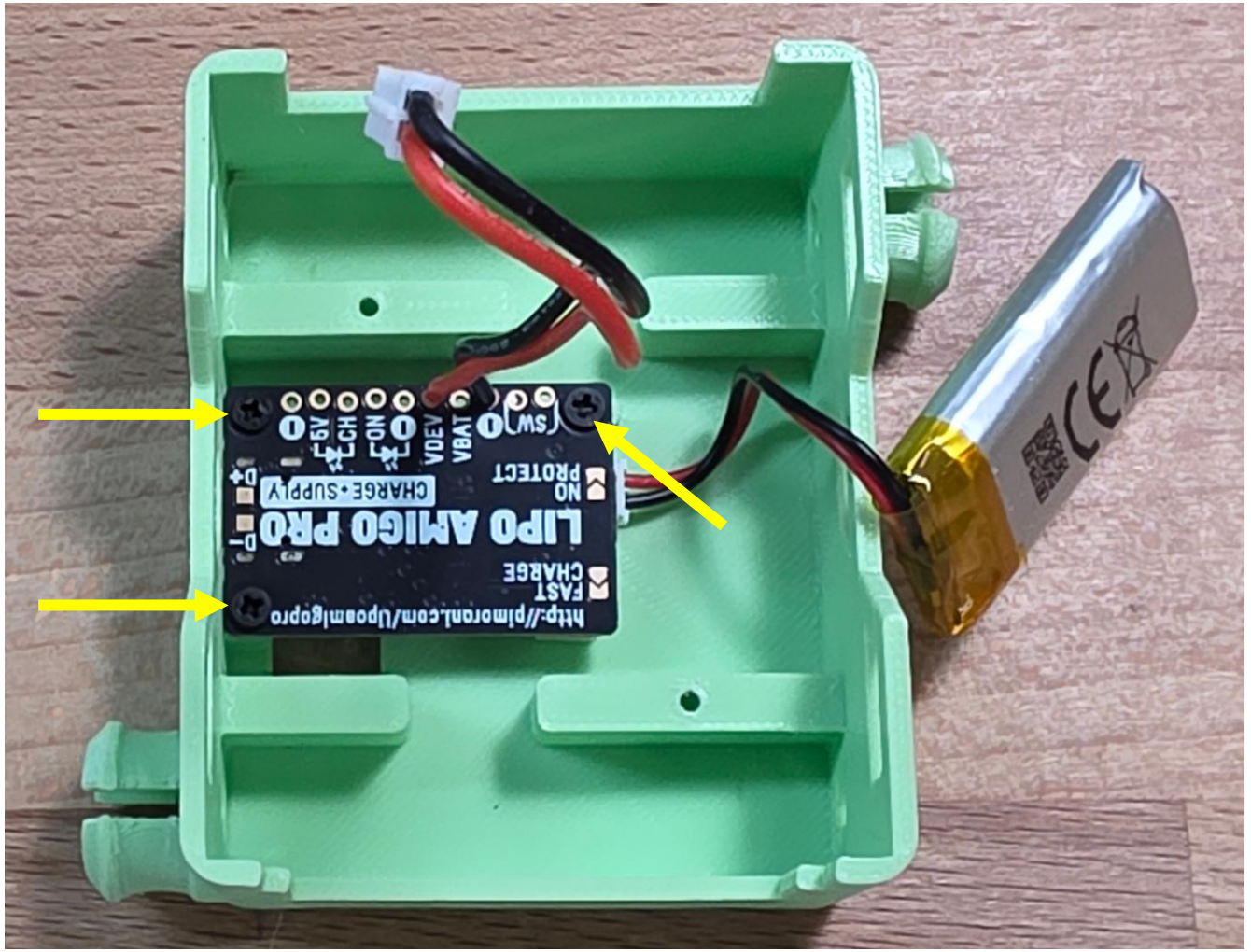


Installing Charging PCBA
and Battery (continued)

Use three M2x4 Screws to secure the PCBA to the chassis.

Note: The electronics face down to provide access to the power switch through the window in the bottom of the chassis.

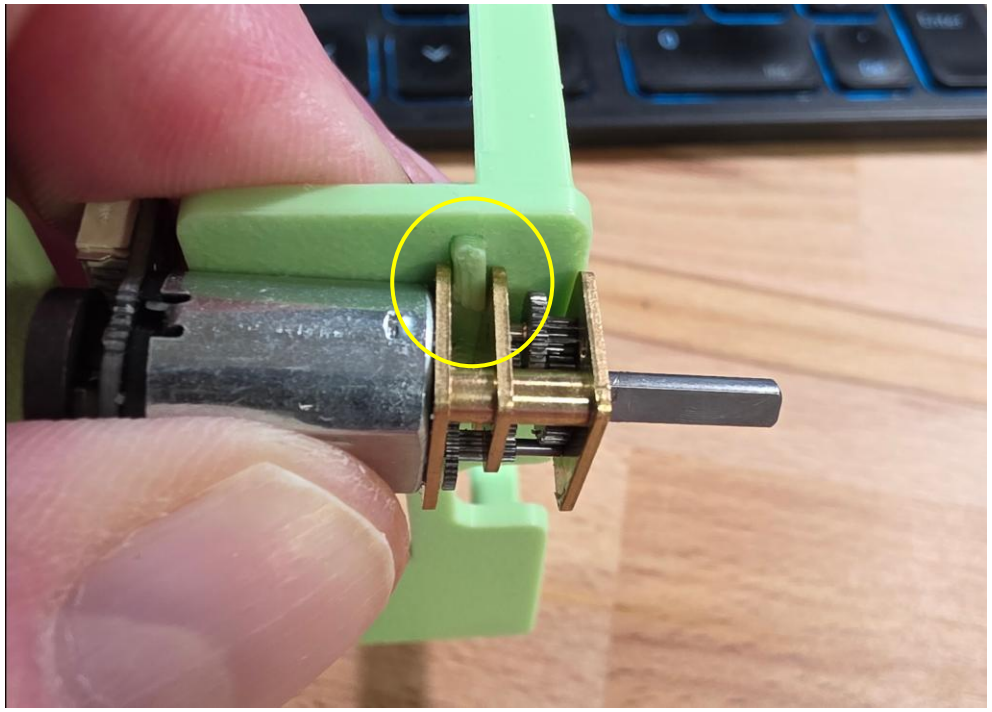


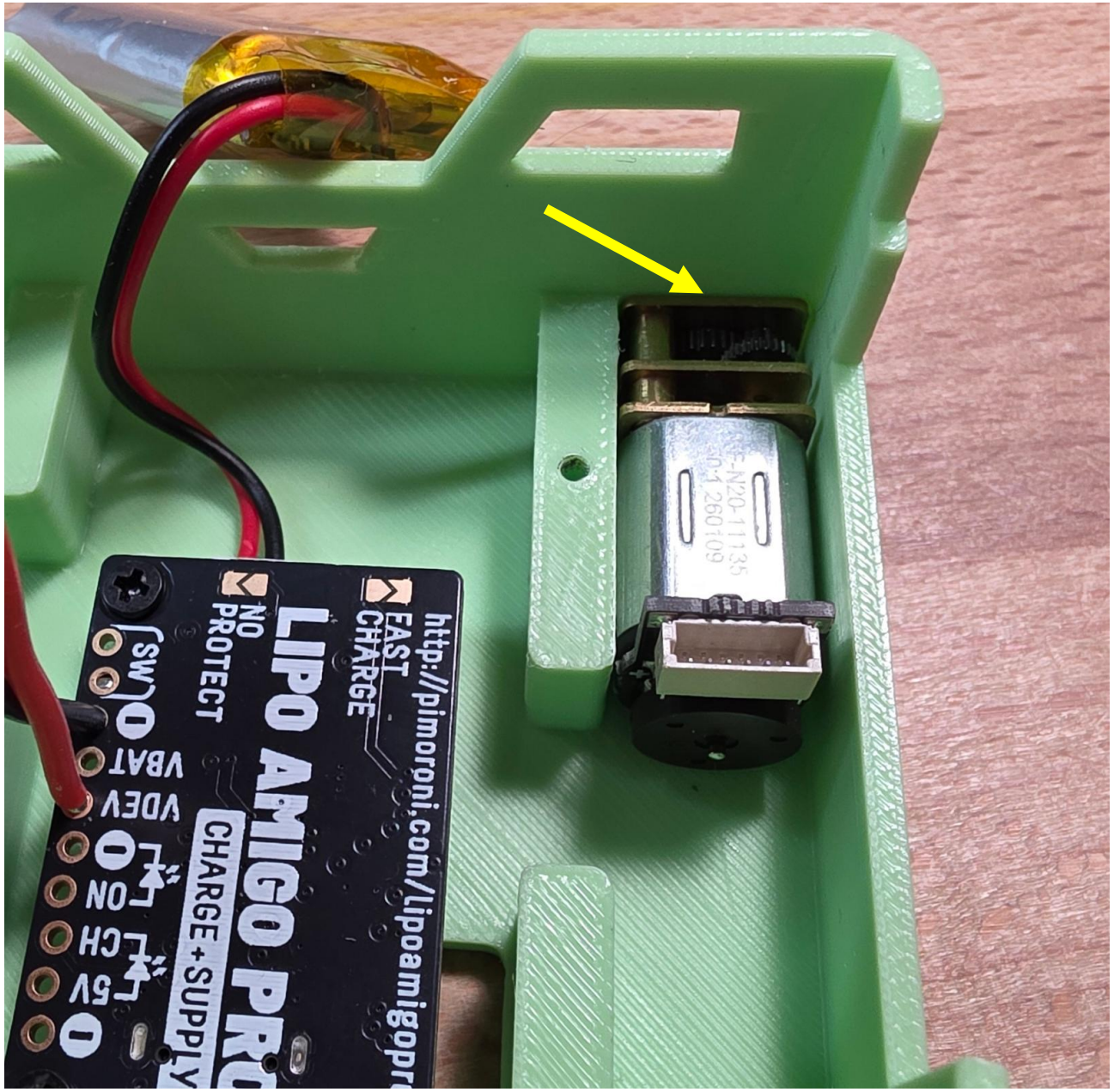


Install Motors

Slide the Gearhead Motors into the Chassis.

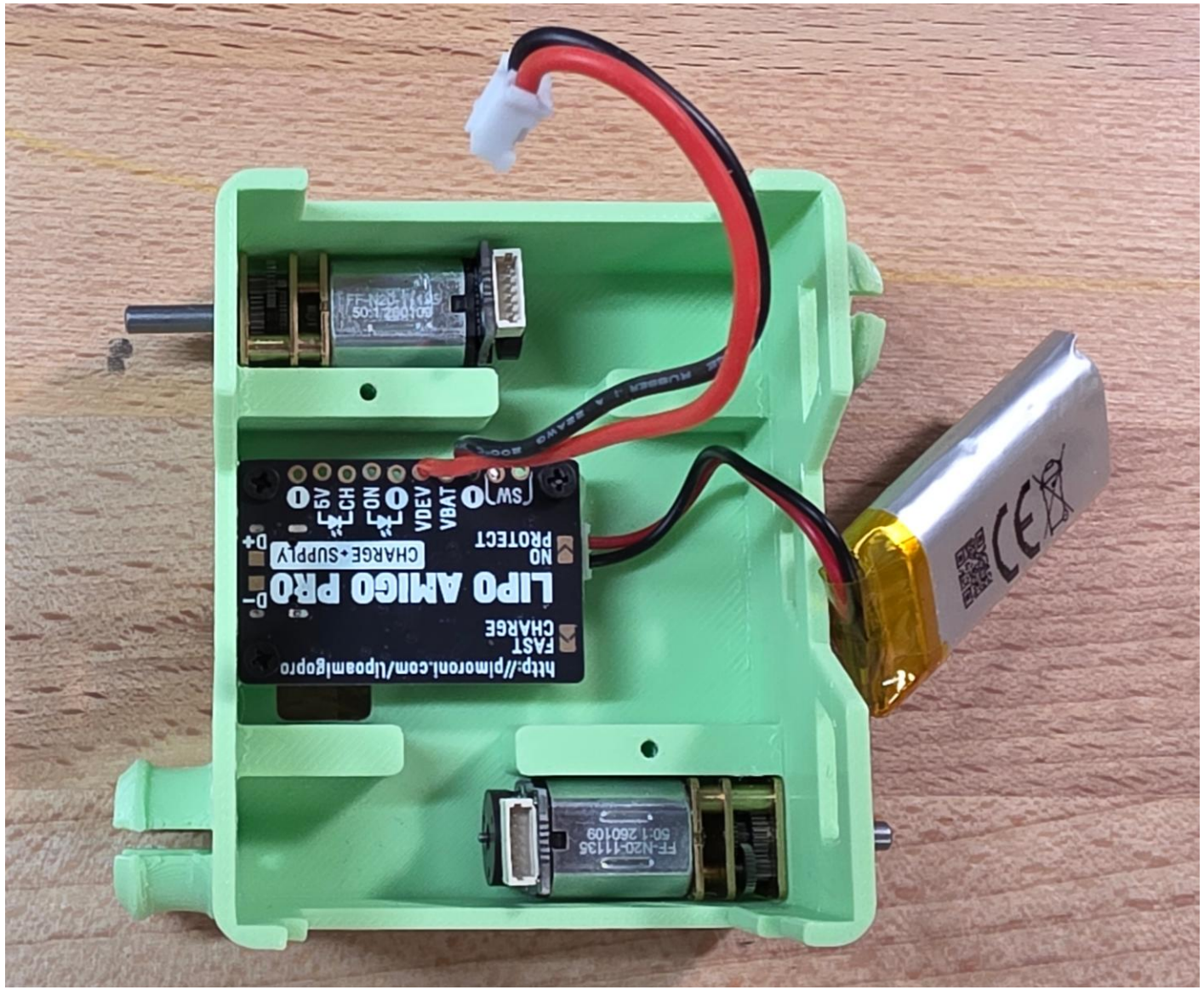
Note: the must be flush with the interior, so the motor lock tabs on the Battery Plate will engage the Gearbox correctly





Install Motors (continued)

Your Chassis should look like this with the motors opposite each other and the power components in place.

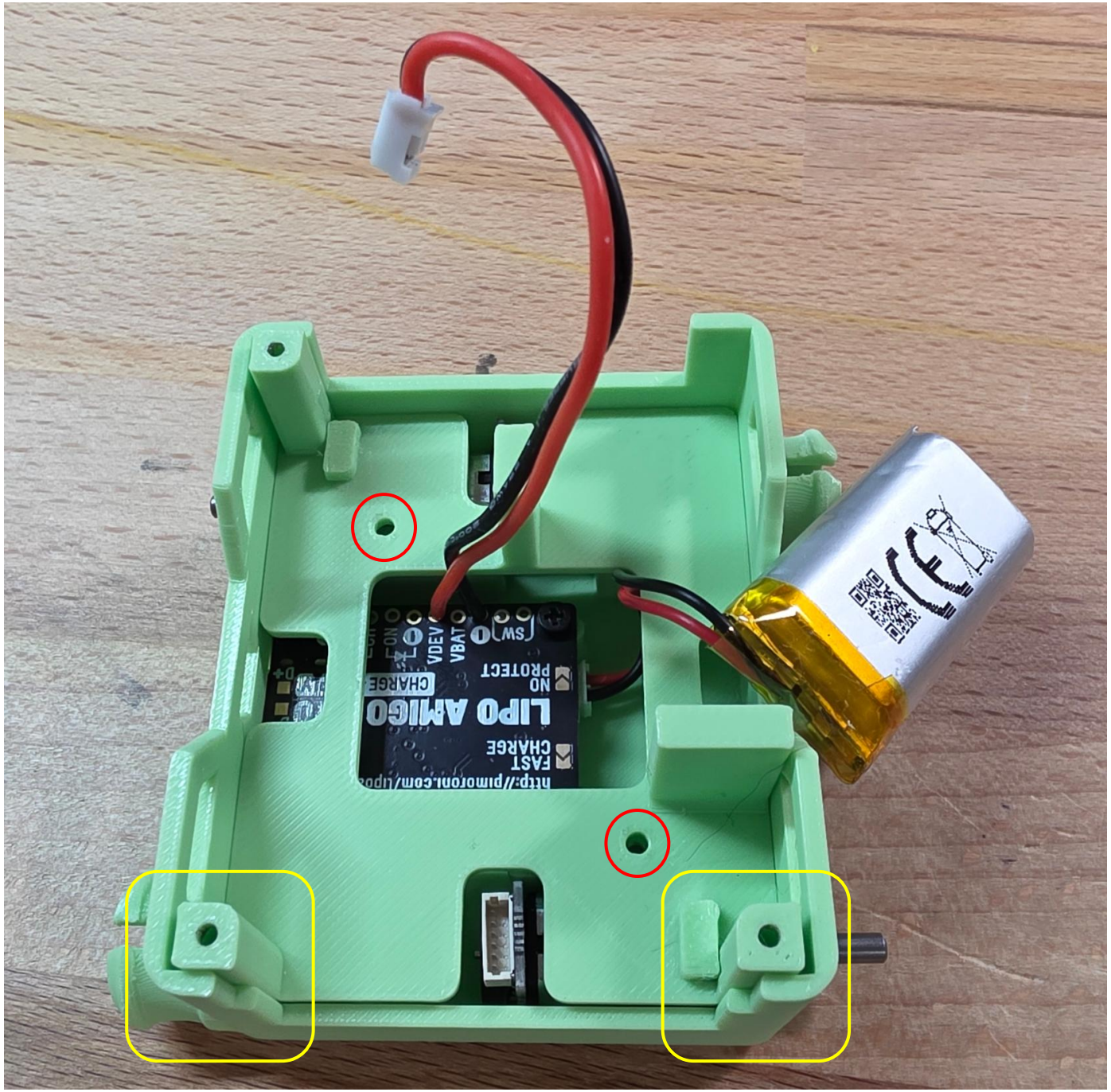


Install Battery Plate

Once the Gearhead Motors are in place install the Battery Plate as shown. Make sure that the Battery and Power wiring go through the center of the plate. Also, that the notched ends in the chassis have two mounting posts near them (yellow boxes).

Note: Ensure the Motors are flush with the interior, so the motor lock tabs on the Battery Plate will engage the Gearbox correctly. You can pull on the shafts as you push the plate into the chassis.

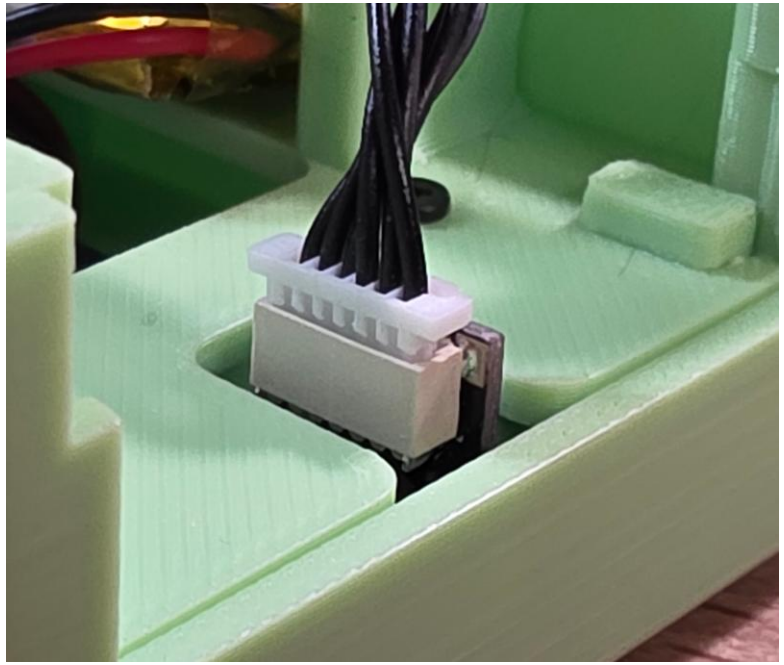
Once the plate is seated in place secure it with two screws (red circles).

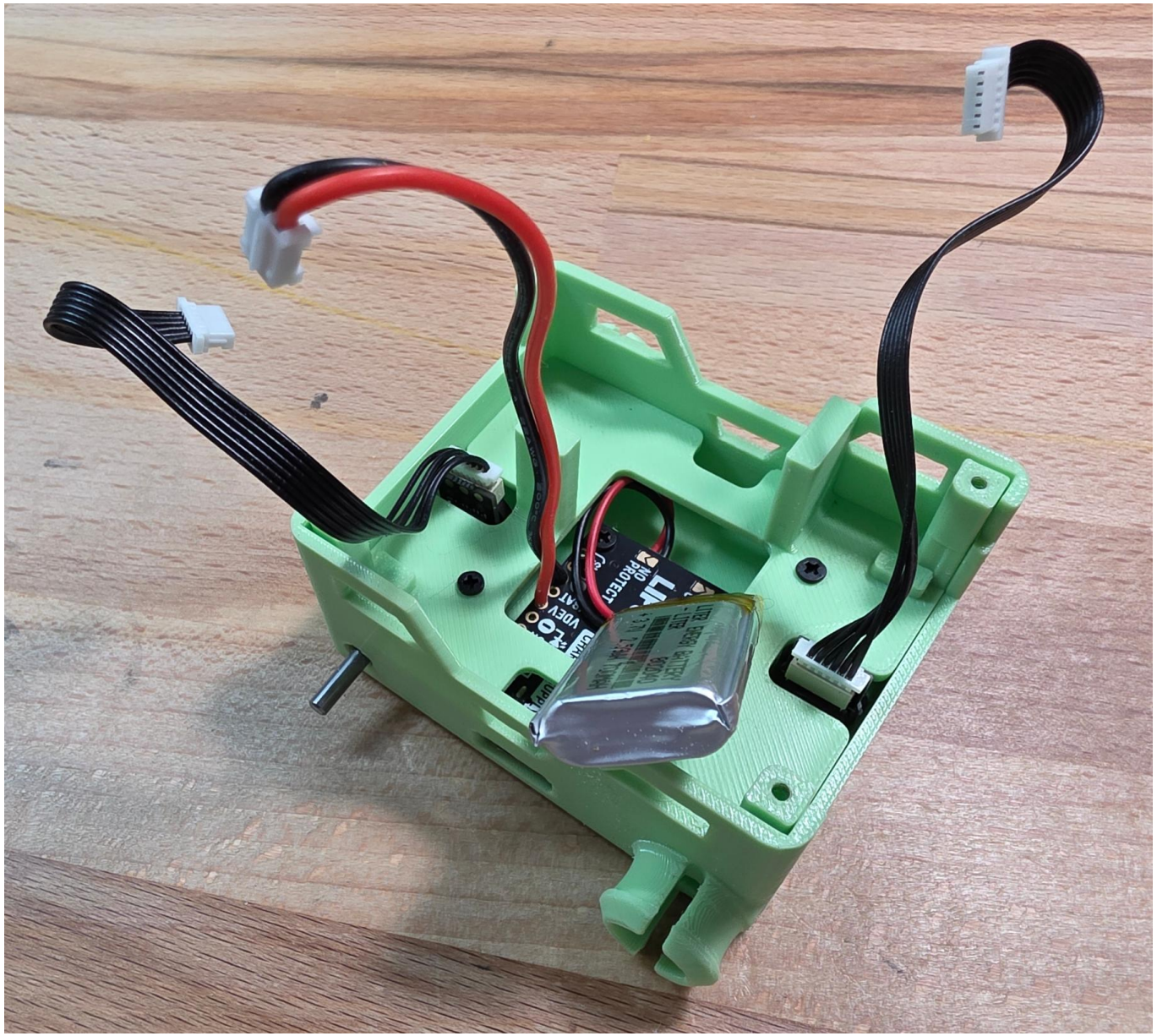


Install Gearhead Motor Encoder Cables

install the Encoder Cables as shown.

Note: These are somewhat fragile and they have very small indexing notches. Use caution as they do not click in place. See fully seated connector below.

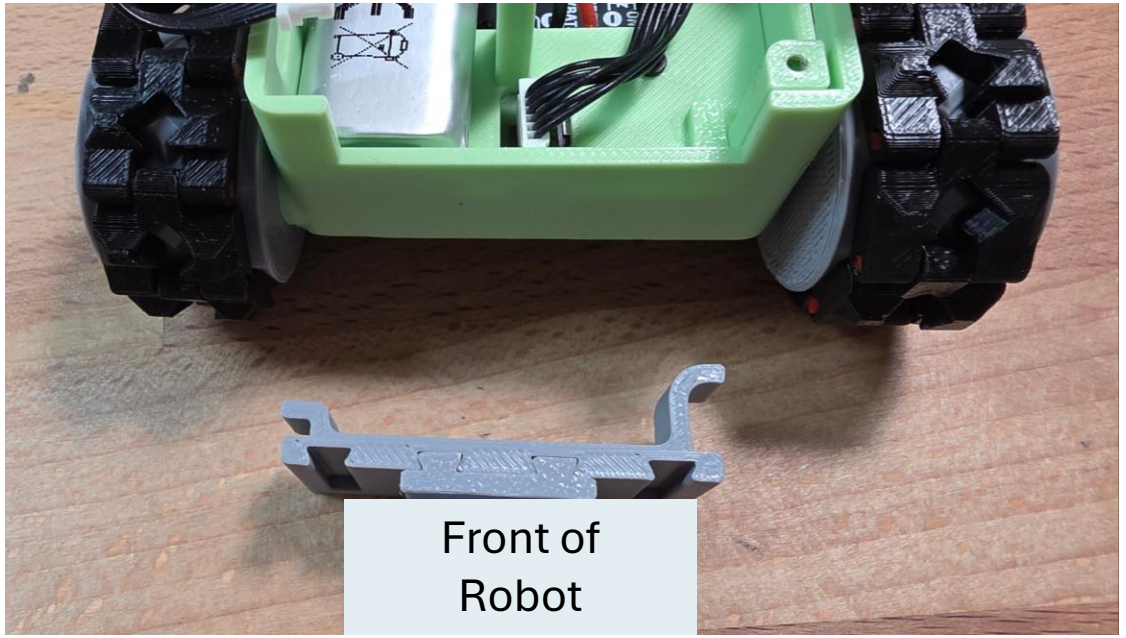


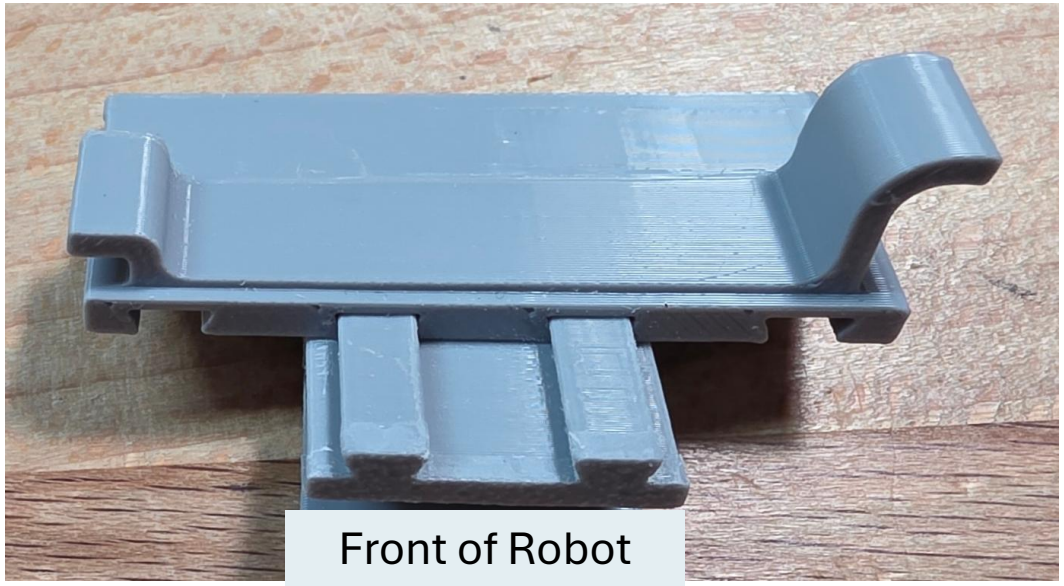


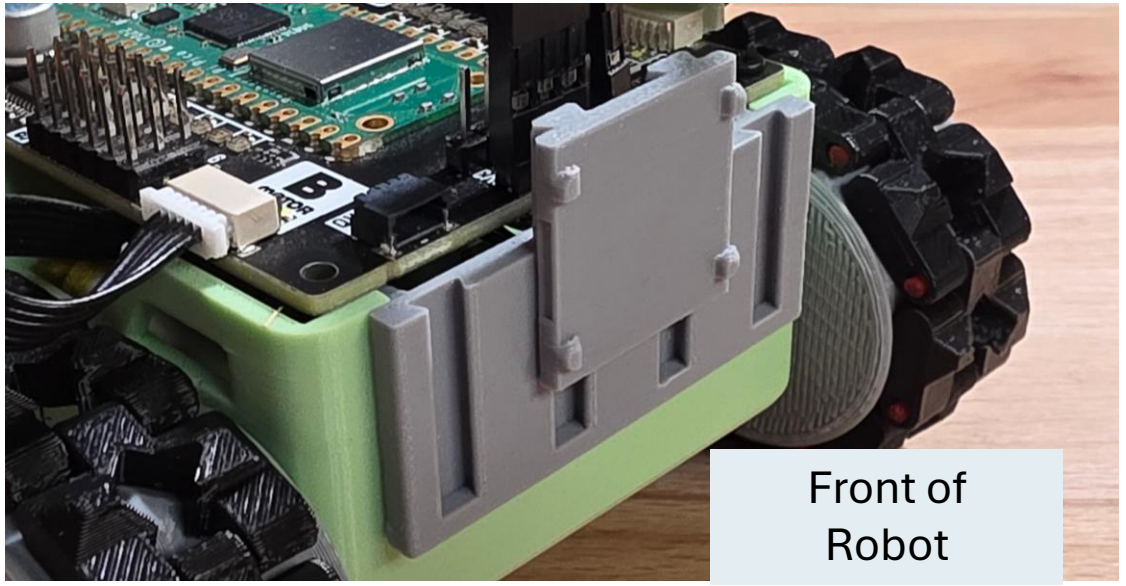
Installing Front Mounting Bracket

Assemble small bracket to the front bracket (see below). When seated they are nearly flush on top (top right).

Slide onto Chassis as shown (bottom right).







Assembling Speaker to
Inventor 2040 PCBA

Plug the Speaker into the connector on the Inventor 2040 PCBA (below)

Turn the PCBA over and mount the speaker as shown by removing the white tape protector and setting in place.





Installing Inventor 2040
PCBA

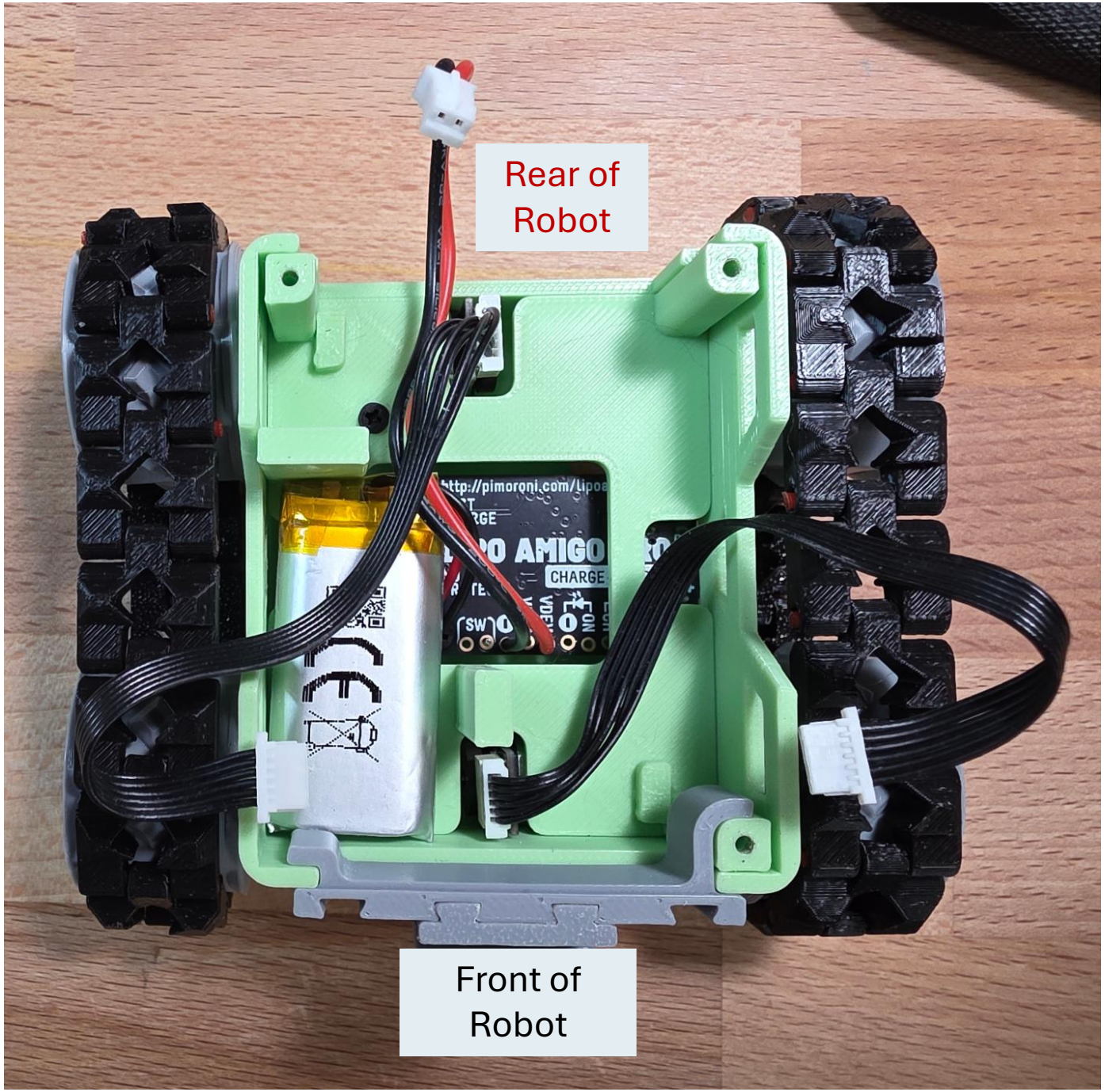
Ensure the battery is placed as shown.

Ensure the cables are routed as follows:

1. The motor cable at the front of the robot is routed to the RH in this view (driver's side from an automotive perspective).

2. The rear motor cable is routed to the LH (passenger side from an automotive perspective).

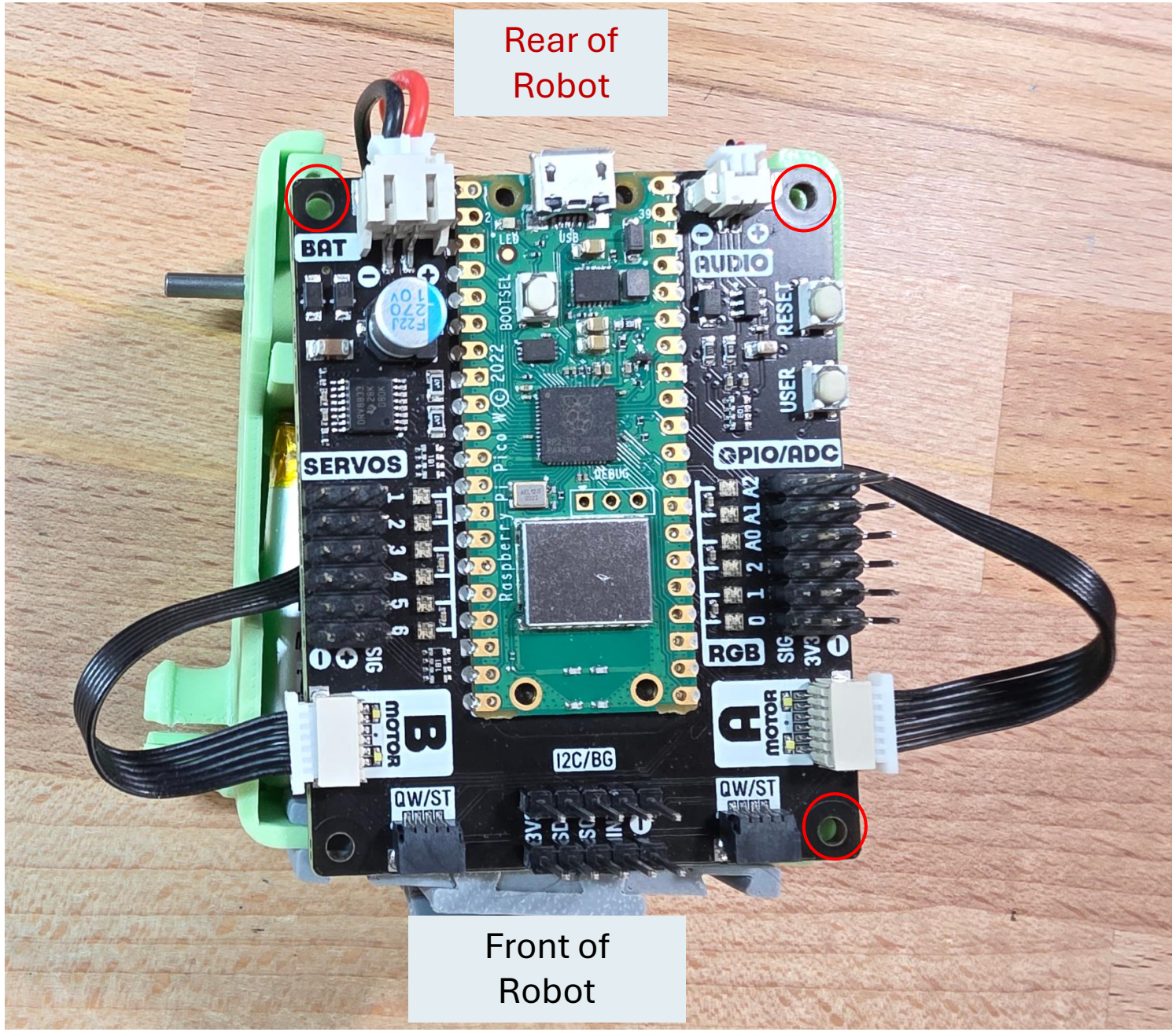
3. The power wiring is routed towards the rear of our robot.



Installing Inventor 2040 PCBA (continued)

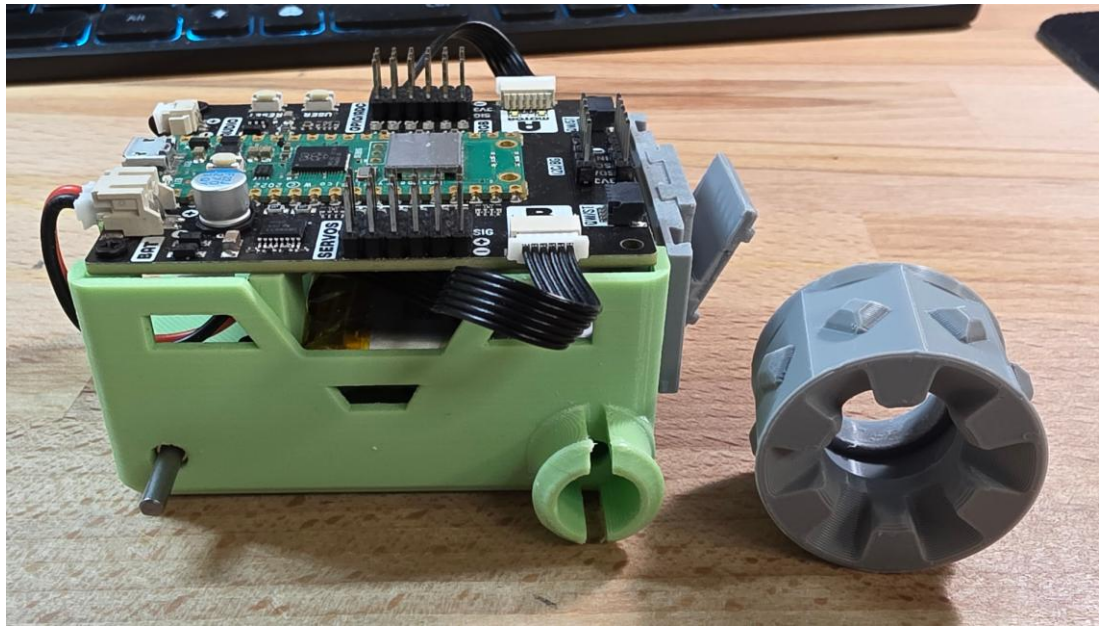
Connect the power and motor cables to the Inventor 2040 PCBA.

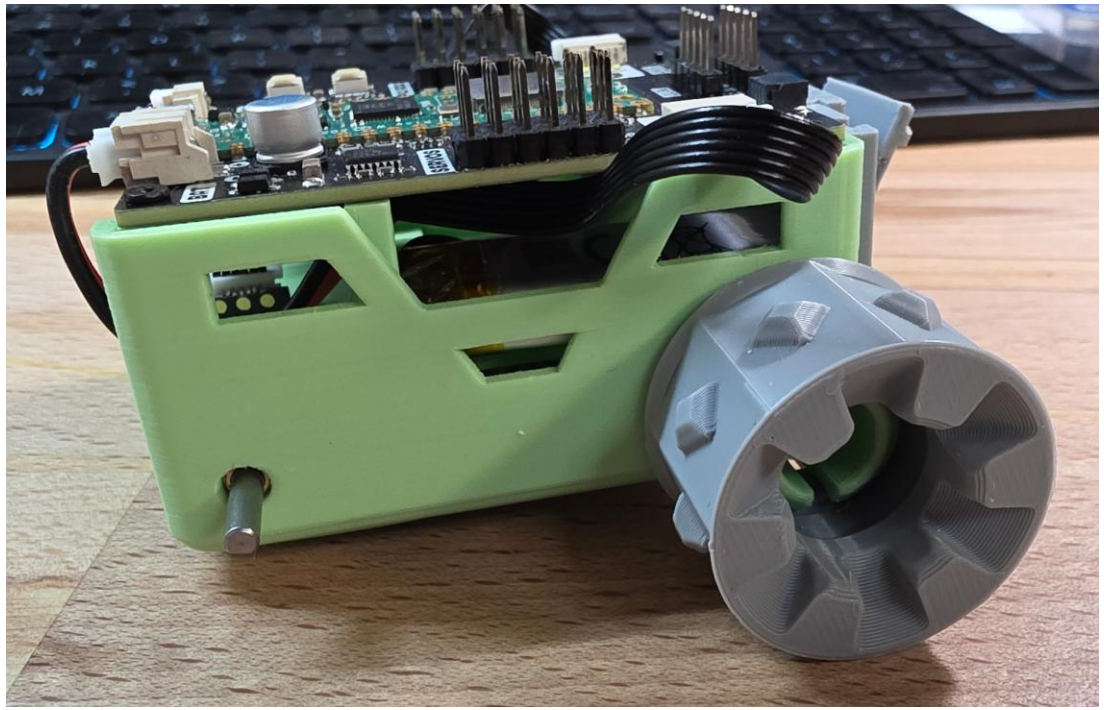
Secure the PCBA to the robot with 3 screws (red circles).



Install Slave Wheels

The Slave Wheels require a bit of push and turn to pop on. Once on the should spin freely without binding.



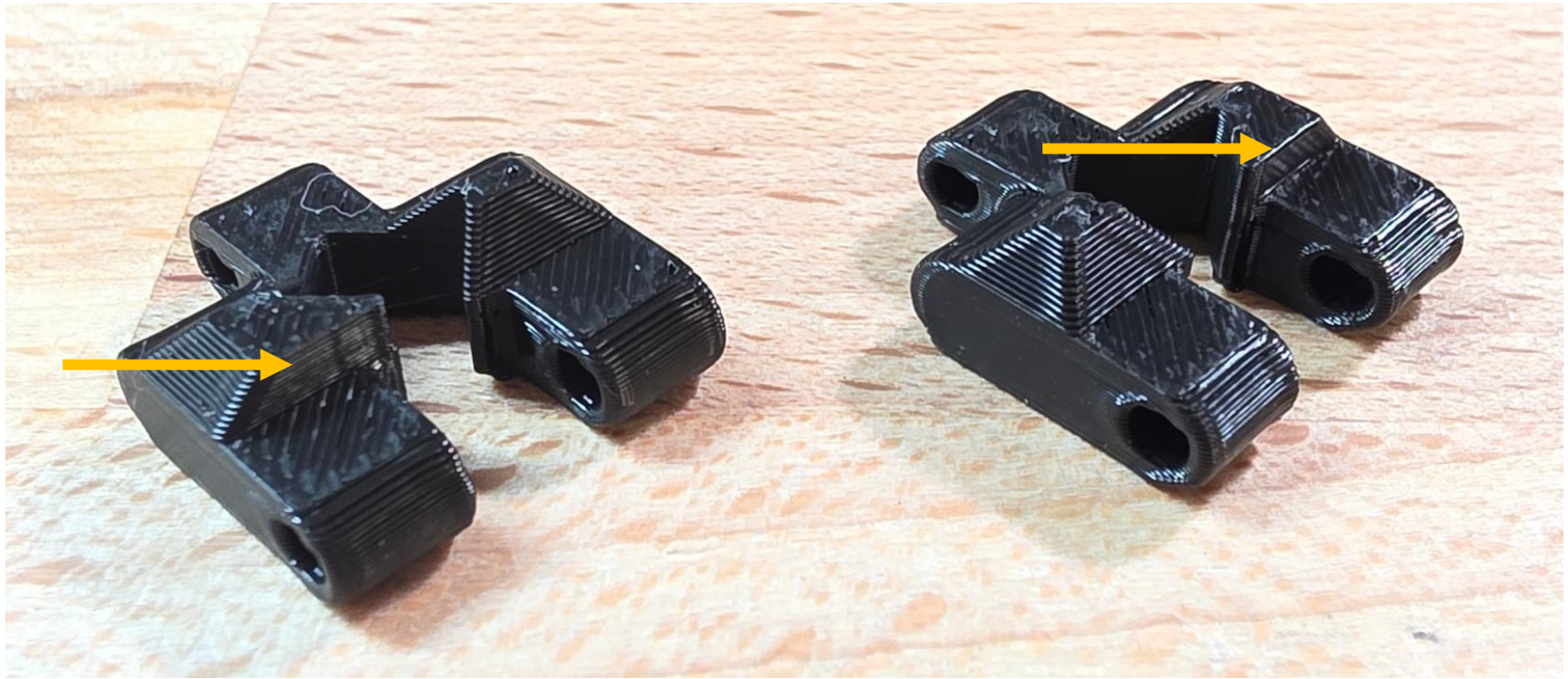


Assembling Tank Tracks

The Tank Tracks are made of 16 segments per track. They use PLA filament as their link points. The parts are mirrored so that you can build a set of RH and LH tracks, or a set of alternating staggered tracks.

Included in the Kit are 17 RH, and 17 LH track segments so that you have one spare per type.

Note how the track segments have a squared off portion on opposite sides.



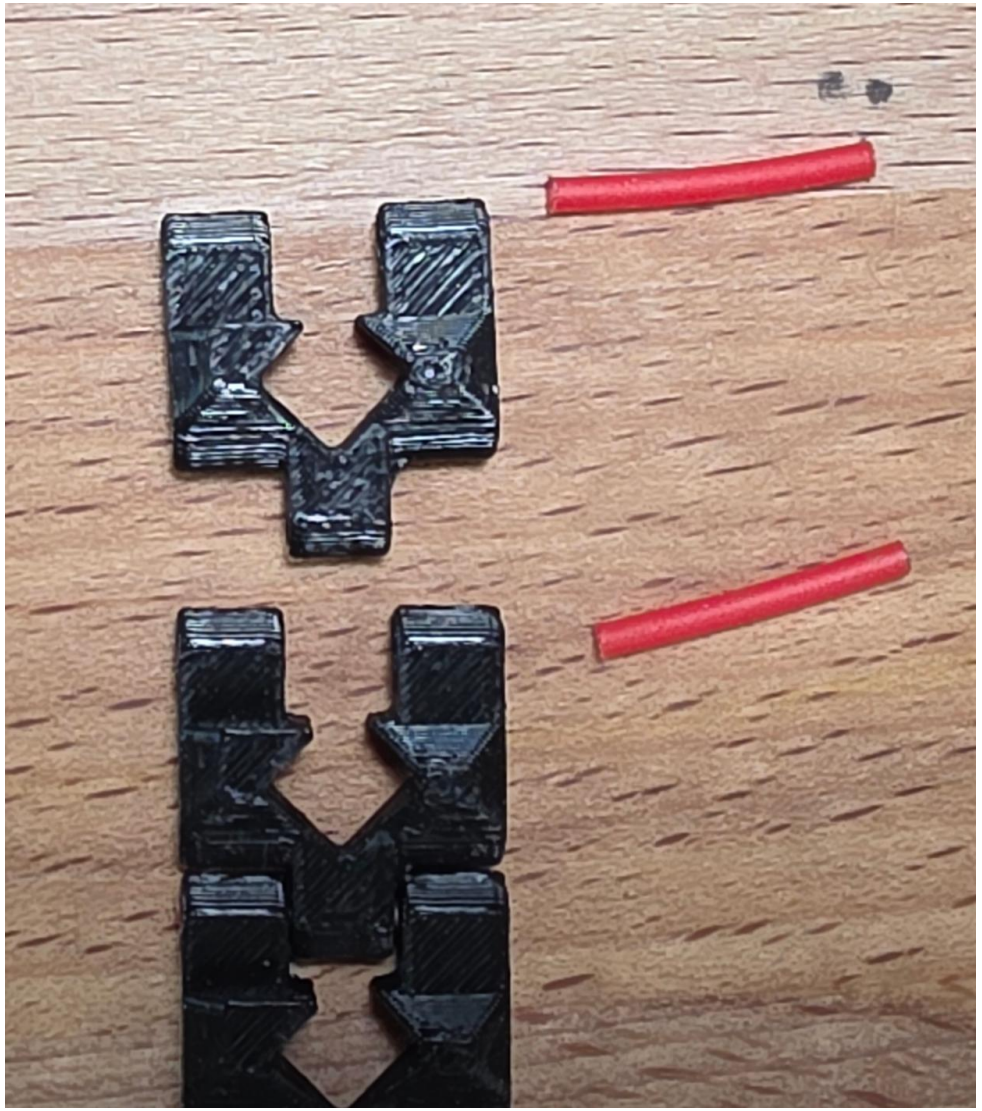
Assembling Tank Tracks (continued)

PLA filament is used as the link points between segments.

Included in the Kit are both yellow and red PLA to use based on preference.

Simply line up the tracks and push the PLA in in until it is flush on the other side. When inserted cut the piece off with and Xacto Knife.

Note: A soldering iron or wood burner can melt the ends to provide a nicer end once cut.

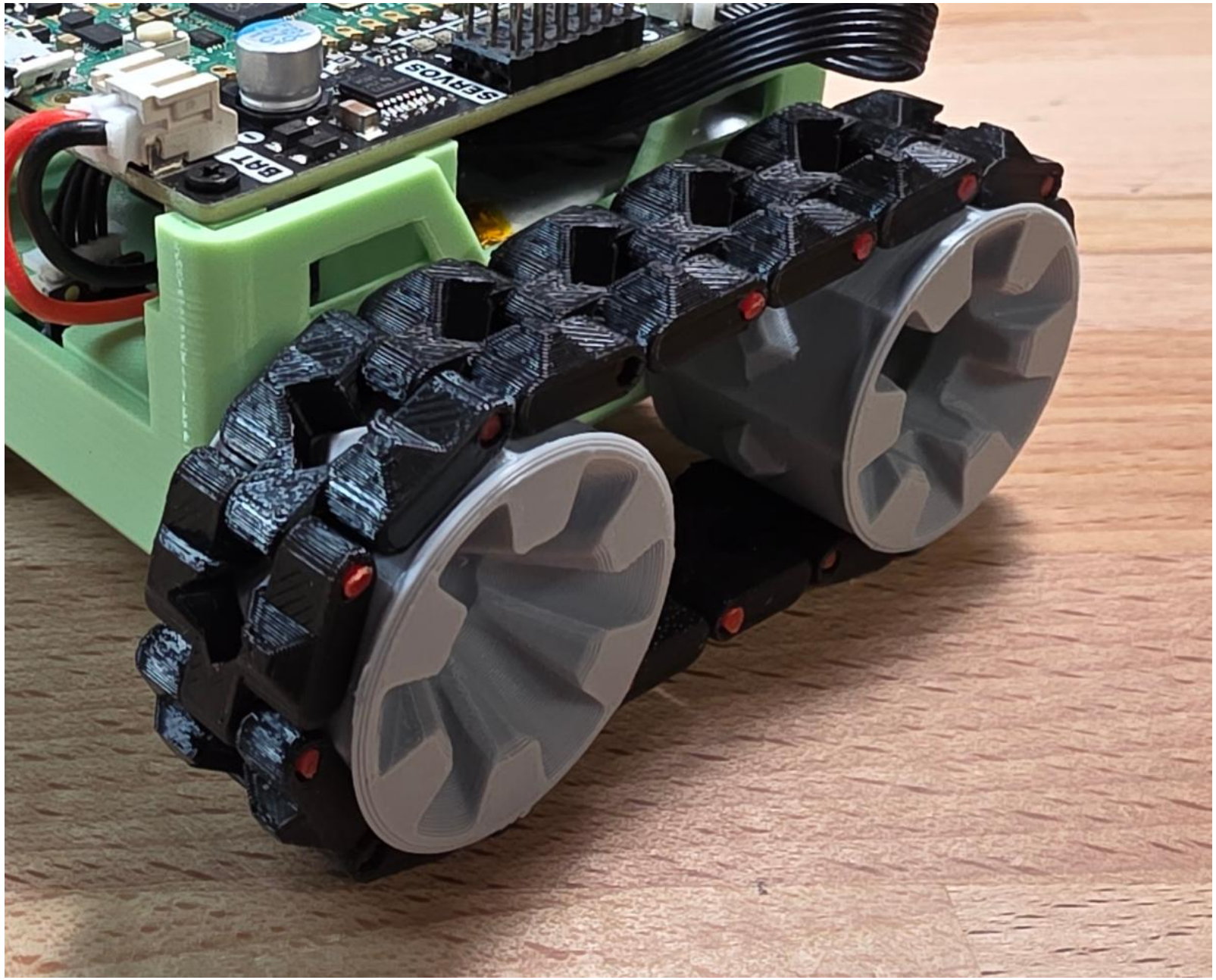


Installing Drive Wheels and Tank Tracks

The Master Wheels and Tank Tracks can be installed in two ways:

1. Install the Wheels first to the D-Shafts, then open the tracks by pulling a link pin. Wrap the tracks around the wheels and reinstall the link pin.
2. Loop the Track around the wheels and push the wheels onto the D-Shaft.

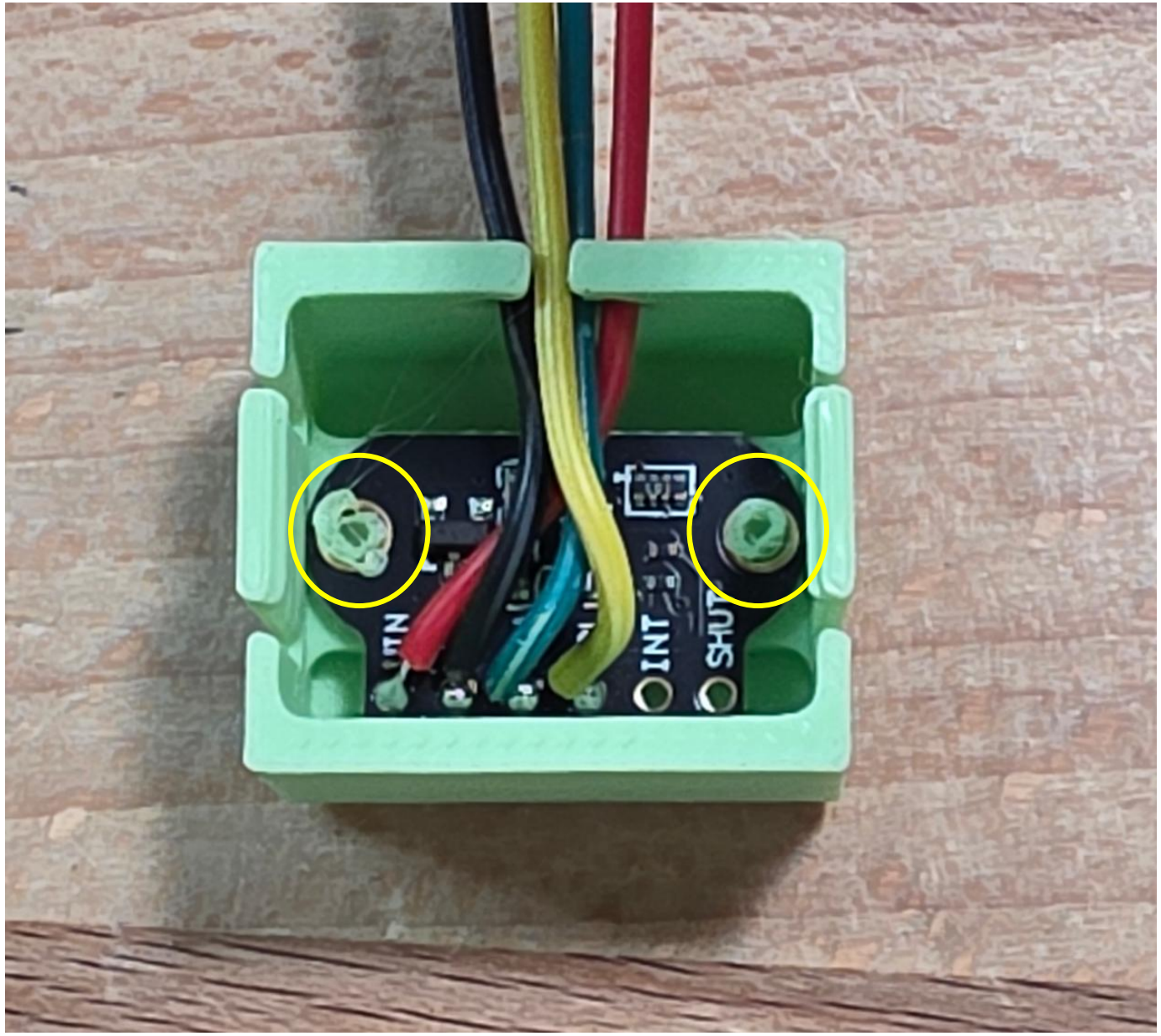
Note, The diamonds on the wheels coincide with the holes in the track. Also, the squared edge should be installed so that the track can grab and help pull the robot along uneven surfaces.



Assembling the VX53L1x
Time of Flight (TOF) Sensor

Insert the VX53L1x into the Housing as shown. Once seated heat stake the pins to secure the sensor. Route the wires up through the housing.





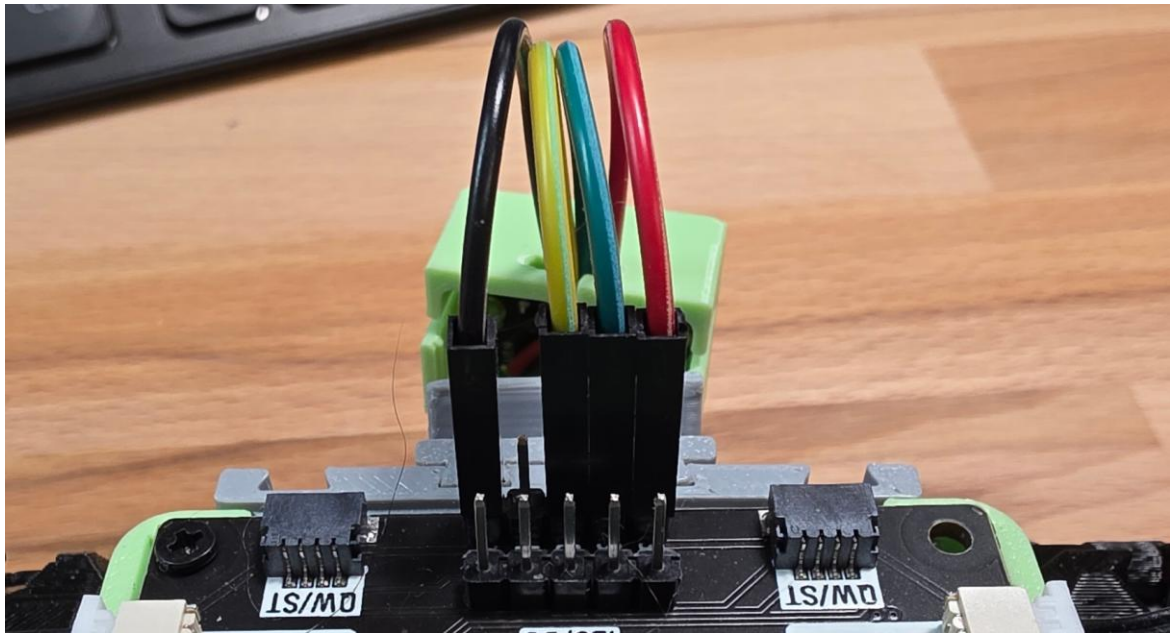
Installing the VX53L1x TOF Sensor Assembly

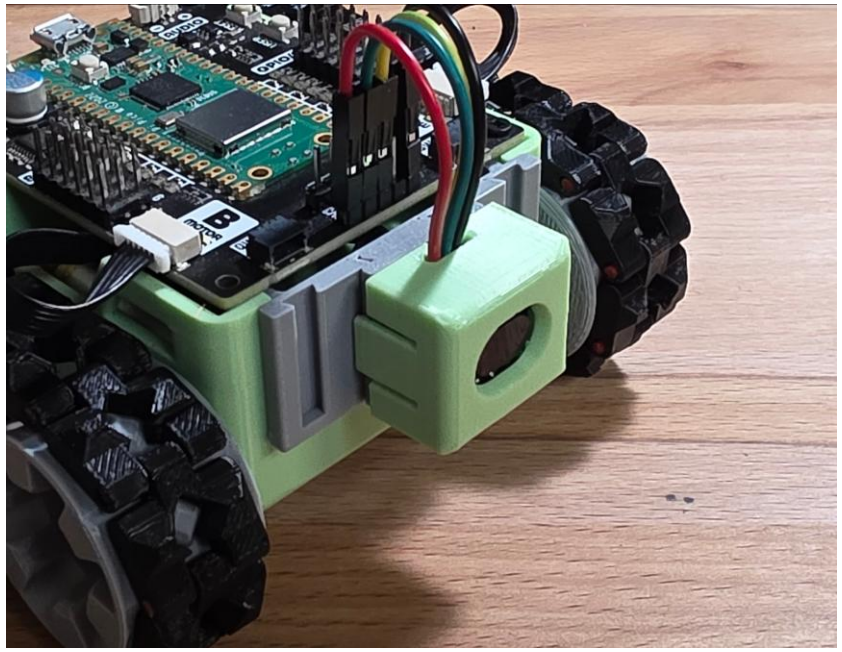
The wiring for the VX53L1x must be connected as shown.

The wiring colors represent:

- Red: VIN
- Black: GND <or> —
- Green: SDA (Serial Data)
- Yellow: SCL (Serial Clock)

Once the wiring is complete snap the housing onto the front bracket.





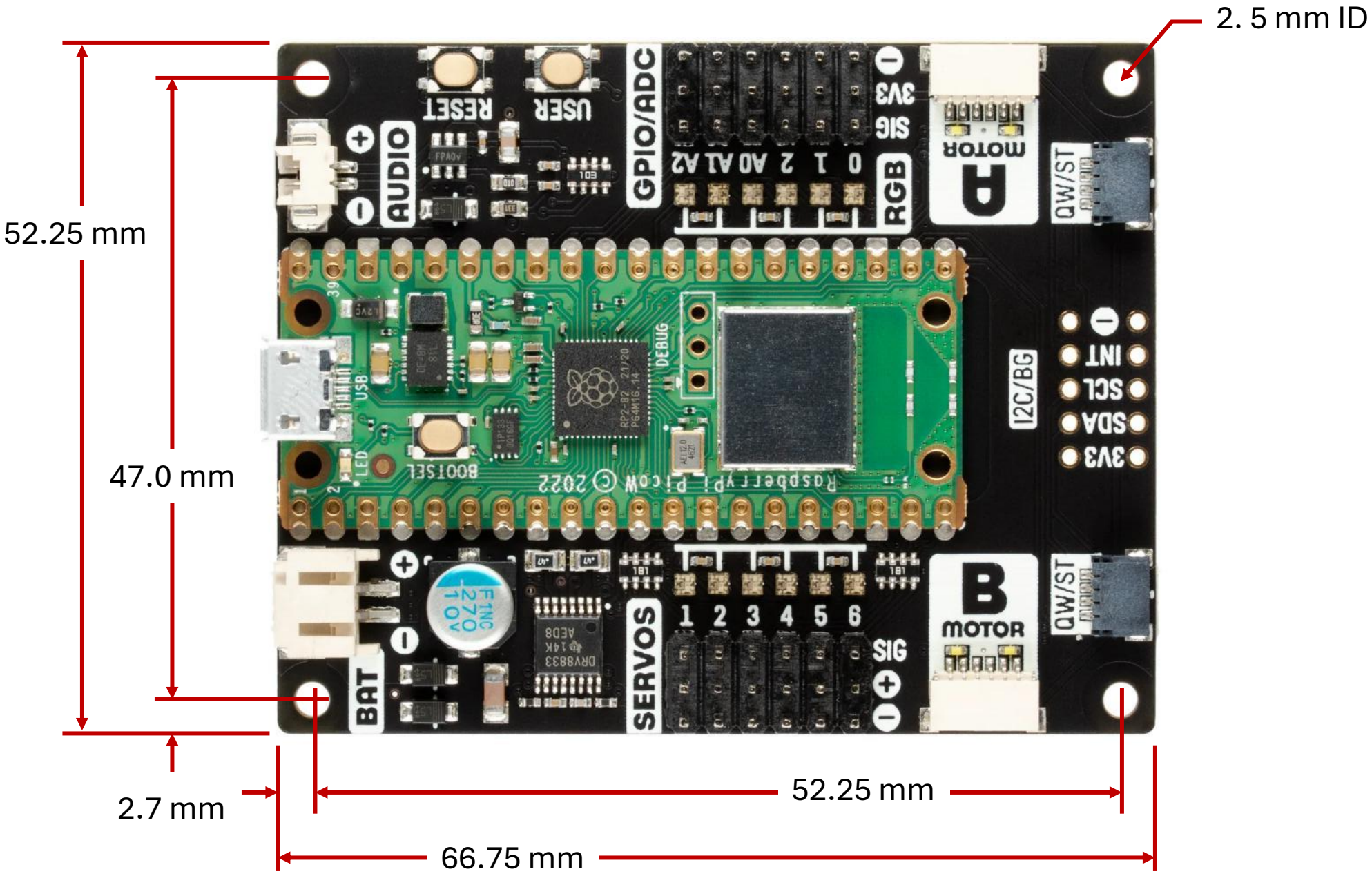
Getting your SMARS to make noise, light the LEDs, connect to a WEB API, demonstrate the motors , and add a simple free ranging robot program

See the:
"Inventor Function Cheatsheet.xlsx"
for Pimoroni's Inventor 2040/2350 W
embedded commands and other info

1. Install Thonny v5.0.0
2. Create a lib folder on Inventor 2040 W
3. Fill out the ssid name and ssid password in the secrets.py file.
4. Upload the secrets.py to the lib folder on your Inventor 2040 W
5. Run the following files to test basic functionality:
    - astronauts_and_iss_r2.py
        - Pulls Astronaut and ISS data from an API on the Web (may need to be restarted a few times)
    - tone_song.py
        - Plays a short tune through the speaker
    - vl53l1x_distance_ranging.py
        - Reports distance in mm
    - rgb_led_test.py
        - Cycles RGB LEDs through the color spectrum then shuts LEDs off
    - motor_position_wave.py
        - Tests motors
    - main.py
        - Upload to SMARS, under battery power the robot will move and avoid objects
6. Happy coding and expanding on your Inventor 2040 W or 2350 W SMARS

Appendix A:
Component Information

Pimoroni Inventor 2040 W Dimensions



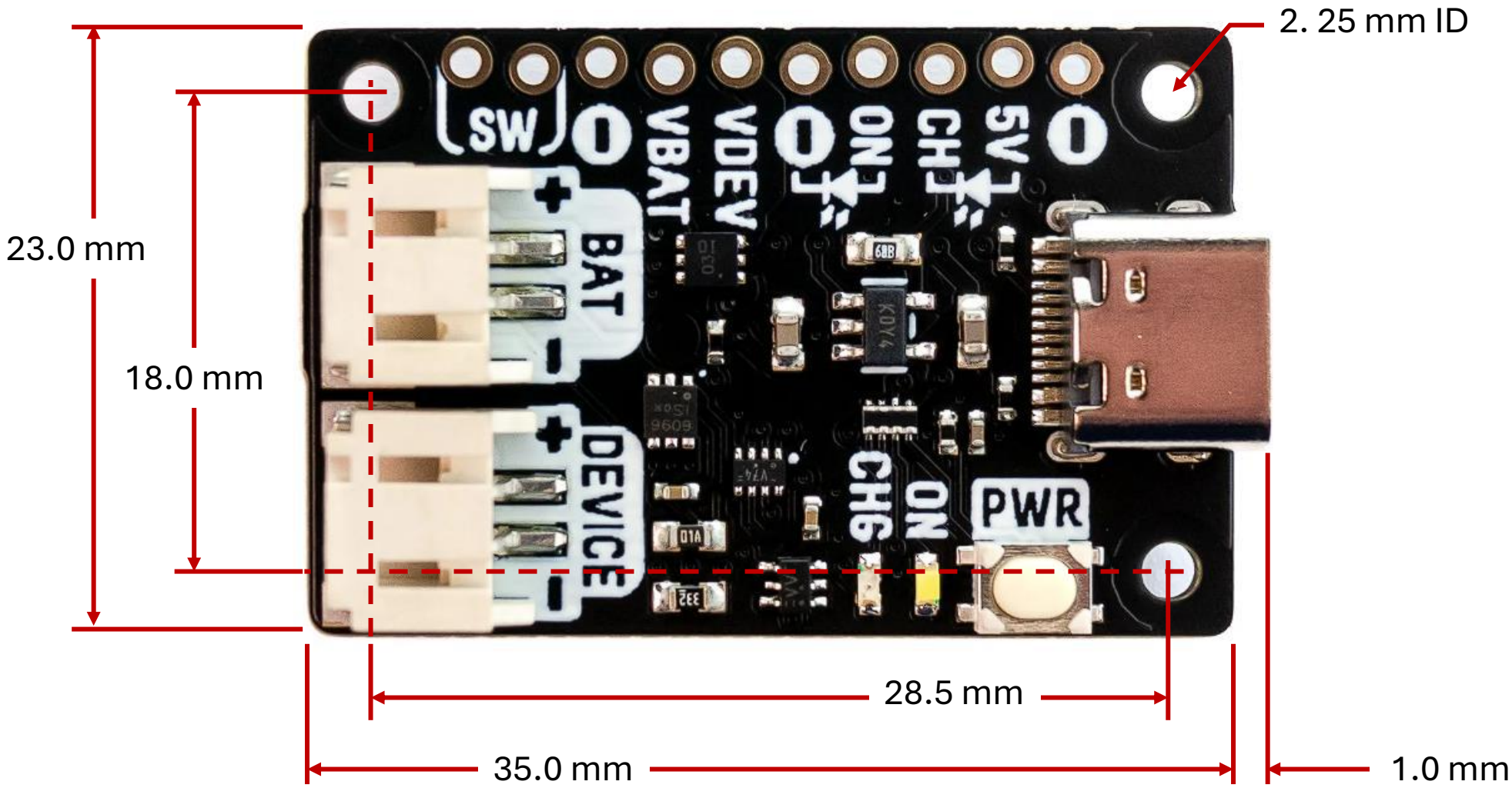
Measurements from Pimoroni:

52mm x 66mm x 12mm (L x W x H).

Mounting holes are M2.5 diameter and 2.7mm in from each edge.

https://shop.pimoroni.com/en-us/products/inventor-2040-w?variant=40053063155795

LiPo Amigo Pro Dimensions



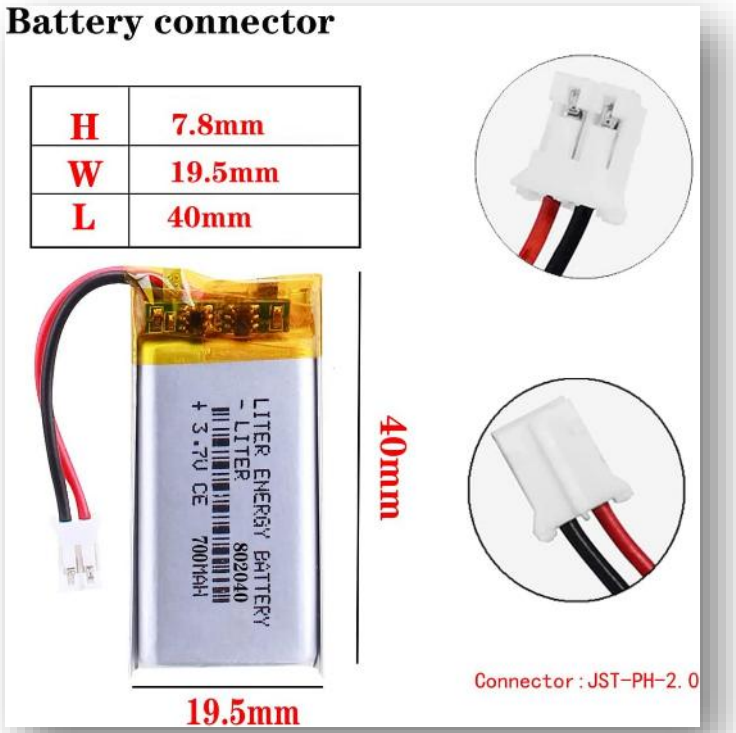
Measurements from Pimoroni:

LiPo Amigo:
26 x 21 x 7 mm

LiPo Amigo Pro:
36 x 23 x 7 mm
(including connectors)

https://shop.pimoroni.com/products/lipo-amigo?variant=39779302539347

Liter 3.7V 700mAh 802040 LiPo Battery Dimensions





https://amazon.com/

Liter 3.7V 700mAh
802040 LiPo
Battery

Capacity:700mAh

Voltage: 3.7V

Charge Ending Voltage:4.20±0.03V

Discharge Ending Voltage:2.75V

Max.Charging Current:0.5C (5℃～+45℃)

Max. Discharging Current:0.5C (-20℃～+60℃)

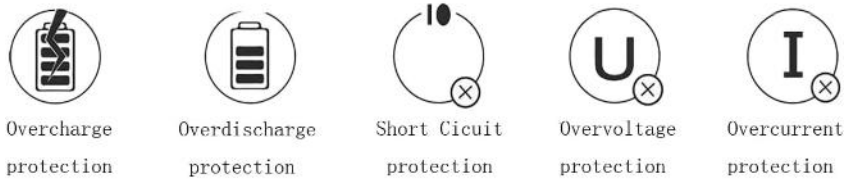
Charge Temperature:0~45℃

Discharge Temperature:-20~+60℃

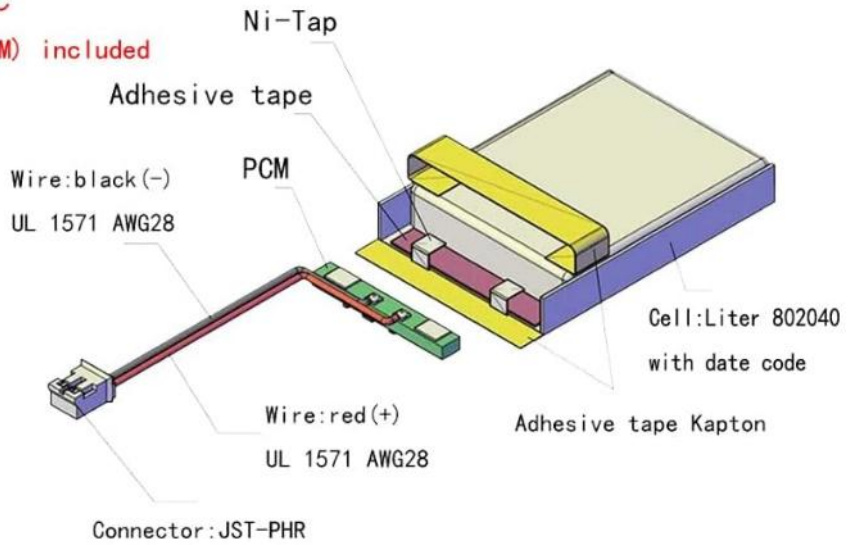
Battery Protection Circuit (PCM) included

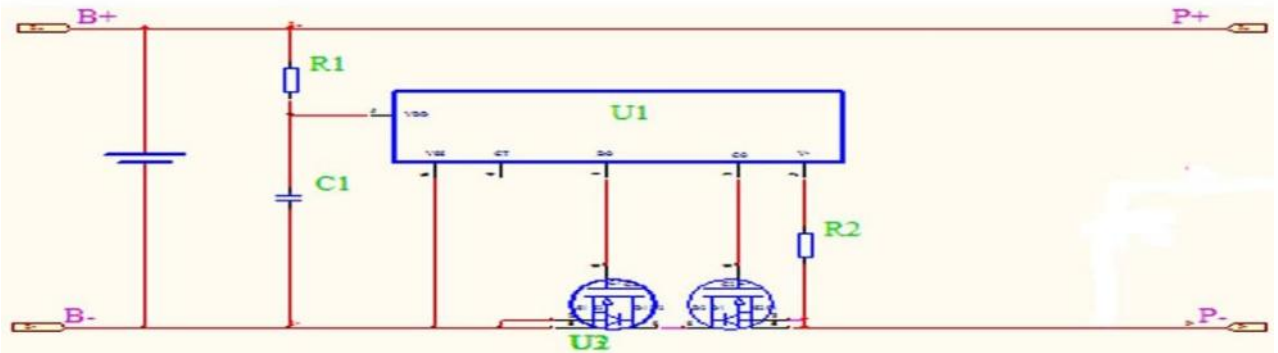
Liter 802040

Multiple Protection









Micro Metal Gearmotor with Encoder (MMME)



One of our popular micro metal gearmotors, equipped with a pre-soldered Micro Metal Motor Encoder (MMME) shim.

We've fitted this motor with a MMME shim that upgrades it to a super smart **quadrature encoder motor**, which uses two Hall Effect sensors and a 6 pole magnetic disc to provide your microcontroller with feedback on how fast and how much your motor is moving.

You can connect the MMME shim to a controller board (like Motor 2040) using a 6 pin JST-SH cable, with no soldering required **(motor cables are sold separately).** Using cables to connect up your motors means they're easy to incorporate into your robotics designs, and straightforward to connect/disconnect if you need to swap out your motors for ones with different gear ratios.

If you need a motor+encoder with a different gear ratio, we also sell MMME shims on their own to solder yourself.

Specifications

This gearmotor is a miniature brushed DC motor with a metal gearbox.

Ratio	Speed @ 6V	Free-run current @ 6V	Stall torque @ 6V (minimum)	Stall current @ 6V	
50:1	420±10% rpm	90mA max	0.4kg. cm	920mA	datasheet

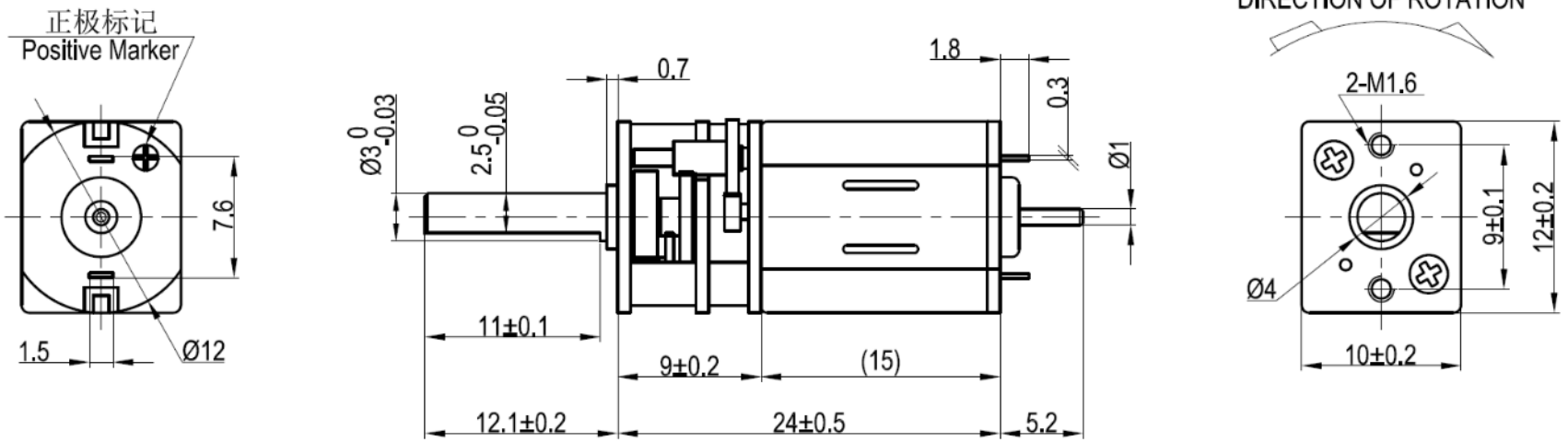
- Long (12mm), 3 mm-diameter D-shaped metal output shaft.

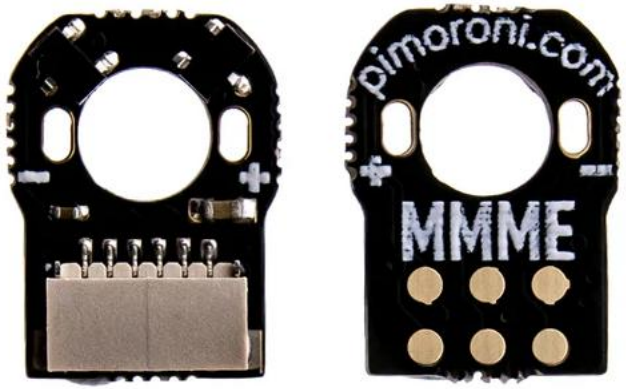
https://shop.pimoroni.com/products/micro-metal-gearmotor-with-micro-metal-motor-encoder?variant=39888597614675

Micro Metal Gearmotor with Encoder (MMME)

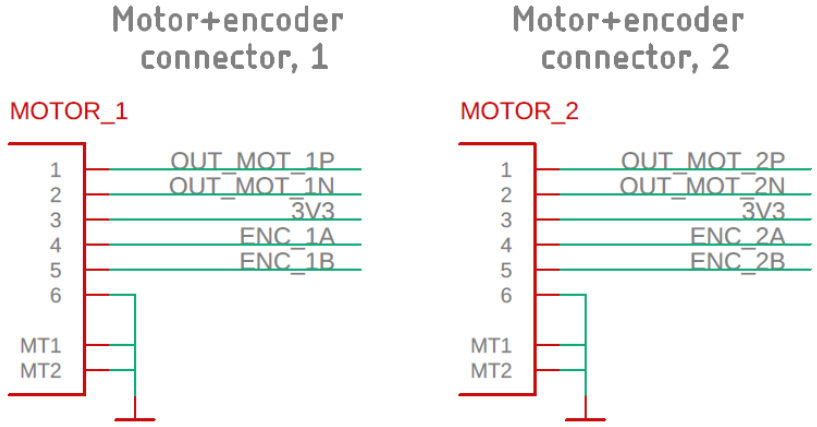
额定电压 V Rated Voltage	6
使用环境温度范围 ℃ Operating Condition Tempertature Range	-10~+60
空载转速 rpm Noload speed	420±10%
空载电流 mA Free-load Current	≤90
负载转矩 Kgf·cm Rated Torque	0.1
负载转速 rpm Onload speed	355±10%
负载电流 mA Roted Current	≤215
堵转电流 mA Stall Current	≤920
堵转力矩 Kgf·cm Stall Torque	≥0.4
旋向 CW/CCW Direction	CW
噪音 dBA Noise (L=300mm)	55 (Max)
轴向间隙 mm Axial gap	0.02~0.3

JL-12FN20-50-06420
Dimensions without Encoder





Connector pinout: M+, M-, 3v3, A, B, GND



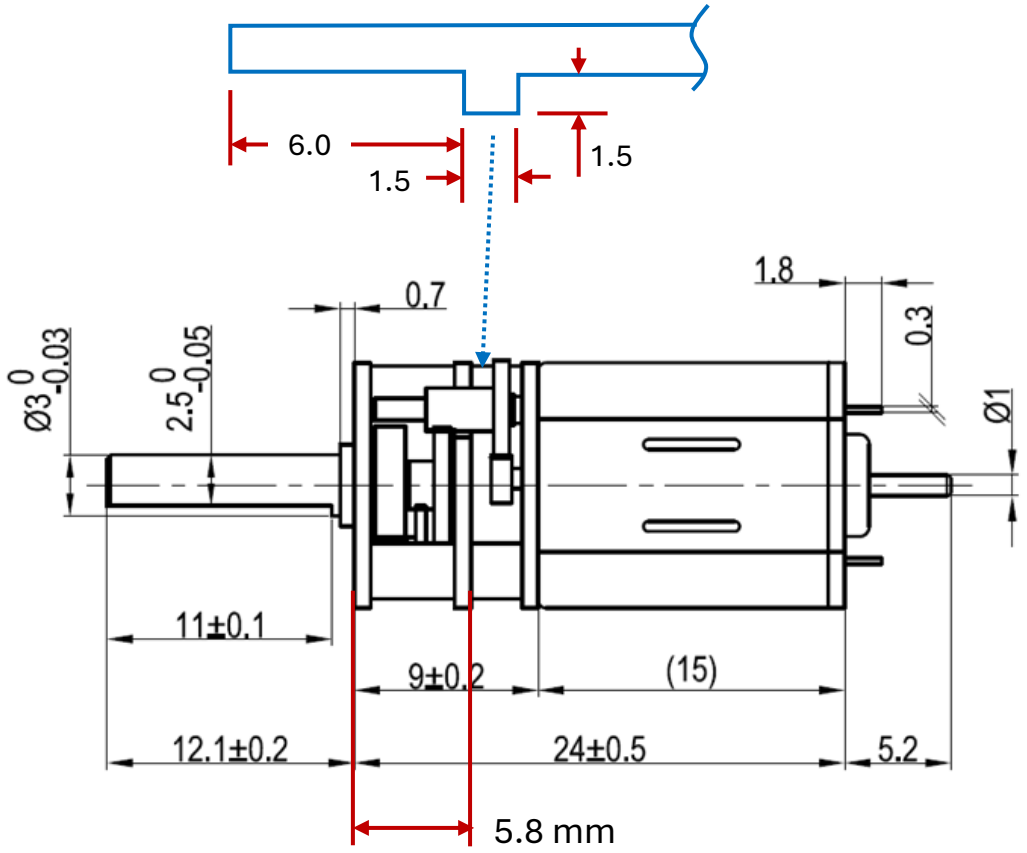
Inventor 2040 W and 2350W Schematics

https://shop.pimoroni.com/products/micro-metal-gearmotor-with-micro-metal-motor-encoder?variant=39888597614675

Micro Metal Gearmotor with Encoder (MMME)



Inventor 2040 Board Mount Motor Lock Feature to lock Behind 2nd Gearbox Plate to Limit Axle Movement Away from Chassis Wall



https://shop.pimoroni.com/products/micro-metal-gearmotor-with-micro-metal-motor-encoder?variant=39888597614675

Adhesive Backed Mini Speaker 8Ω (1W)



A teeny speaker, great for embedded microcontroller projects.

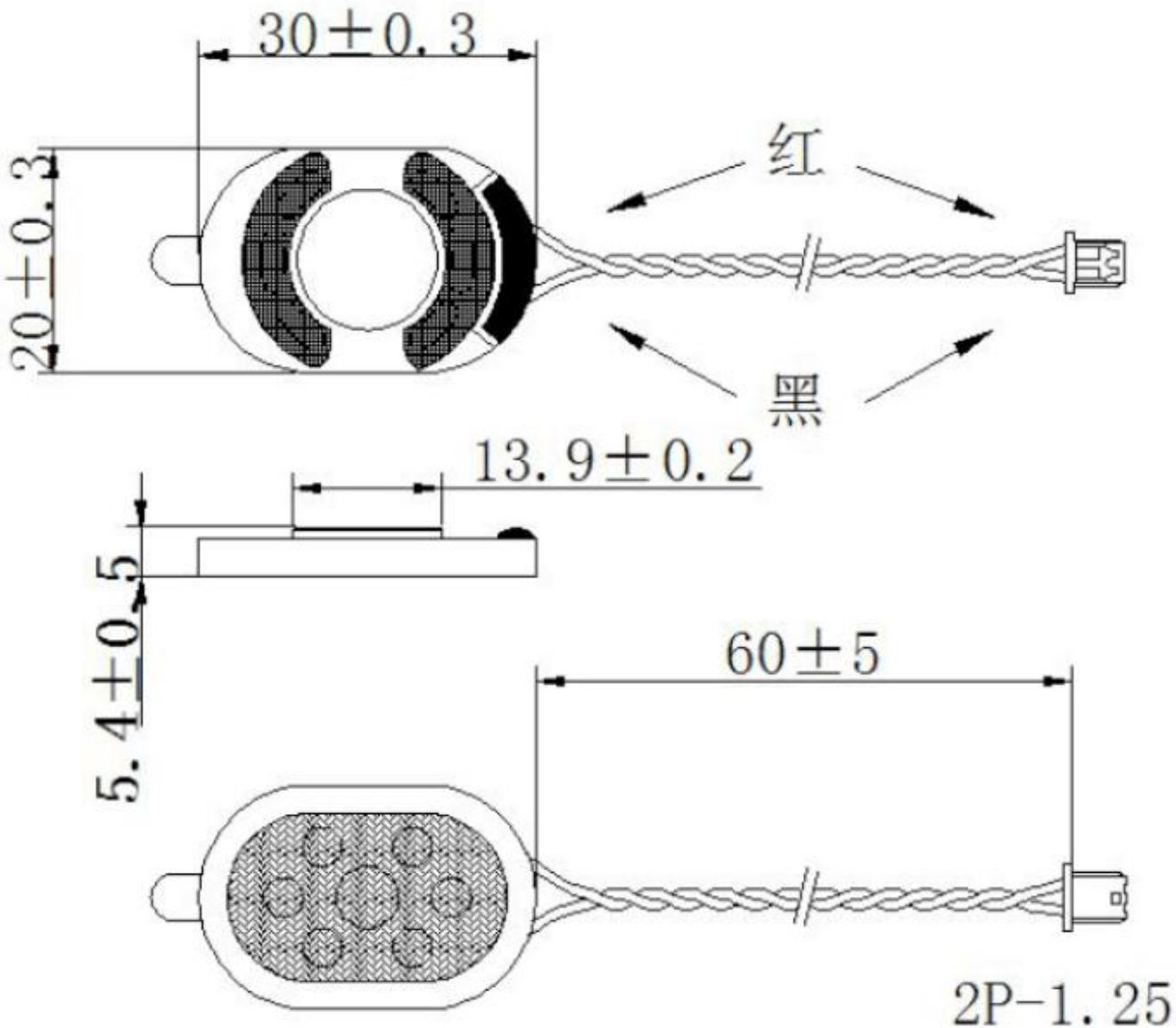
We use this tiny speaker on our Pirate Audio: Speaker and it packs a surprising punch for its size! It has a Picoblade-compatible 2-pin connector, so it's easy to plug into things without the need for soldering or fiddling with screw connectors.

It also works great with Inventor 2350 and Tiny FX!

- Dimensions: 30 x 20 x 4.5mm
- Impedence: 8 ohm
- Power: 1W max
- Datasheet (note our speaker is 4.5mm thick not 5.4mm)

https://shop.pimoroni.com/products/mini-speaker-8-1w?variant=40053581938771

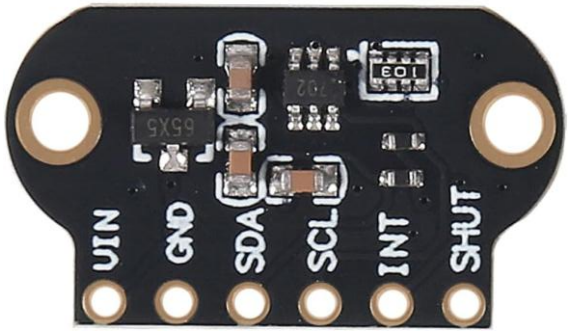
Adhesive Backed Mini Speaker 8Ω (1W)



https://shop.pimoroni.com/products/mini-speaker-8-1w?variant=40053581938771

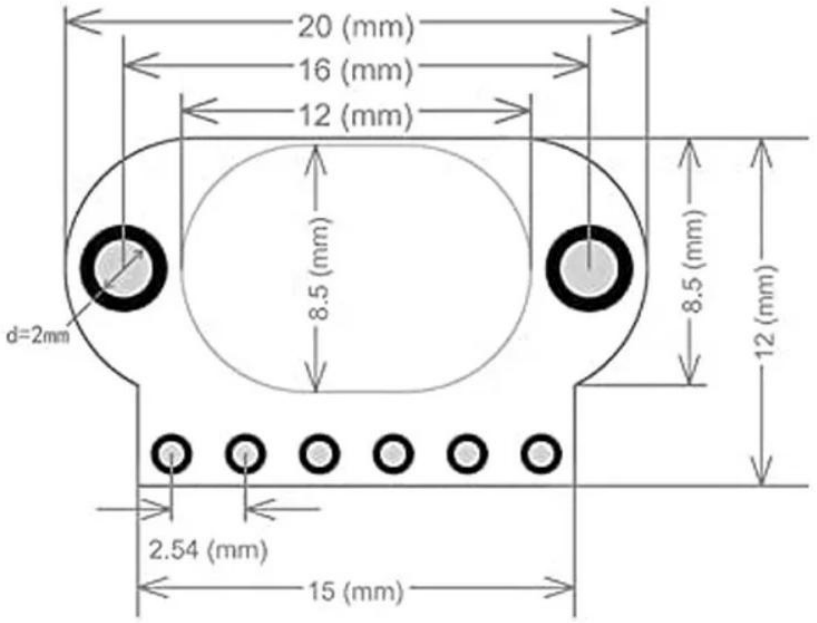
TOF400C-VL53L1x (400cm) Distance Sensor

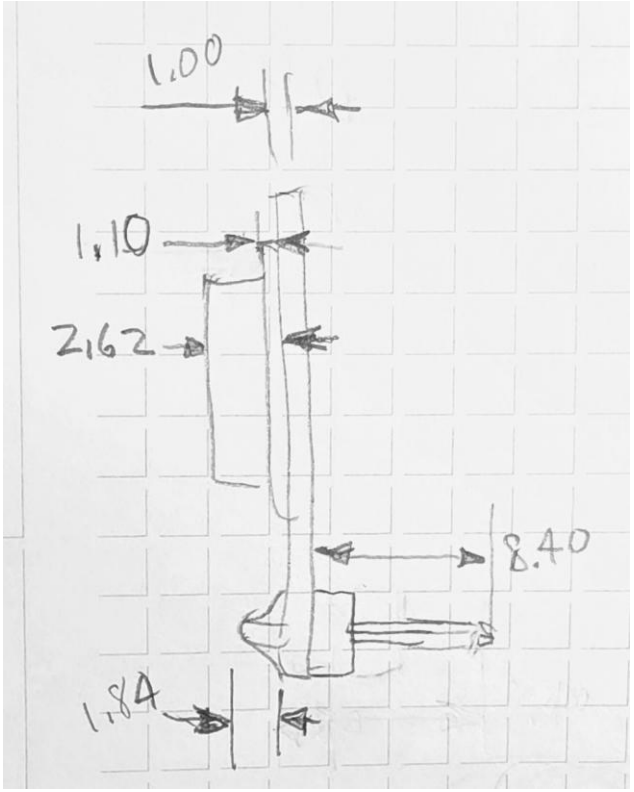




Description:
TOF distance sensor based on the VL53L1x system. allows you to measure the absolute distance regardless of the color and surface of the detected object. Thanks to the application of the IR transmitter emitting light with a wavelength of 940 nm and a set of filters, it offers a range of up to 400 cm with good resistance to interference from ambient light. The module also offers the ability to detect 1D gestures. VL53L1x communicates via the I2C interface, and the user has a friendly and easy-to-use API (Application Programming Interface) available in C language.

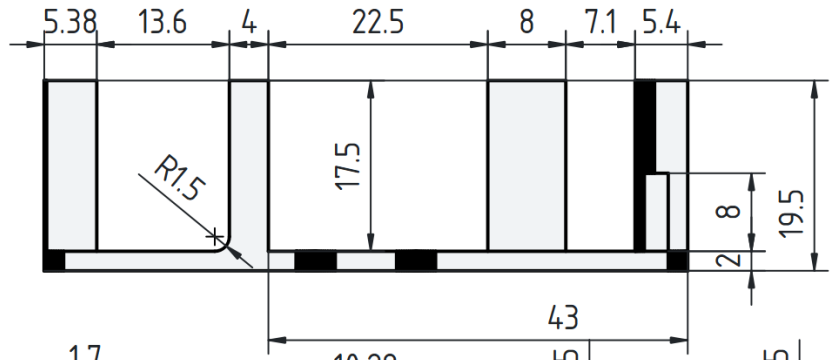
Range: 4 to 400 cm
Dead zone: 0 to 4 cm
Interface: I2C
Programmable I2C address (default 0x52)
Viewing Angle (FOV): 27°
Ranging frequency: 50 Hz
1D gesture detection capability
Protective cover
Two M2 mounting holes
Power supply: 3 to 5 V
Current consumption: 40 mA
Working temperature: -20 to 70°C

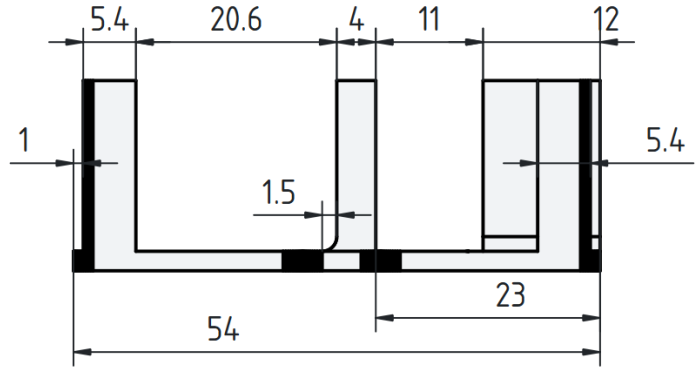


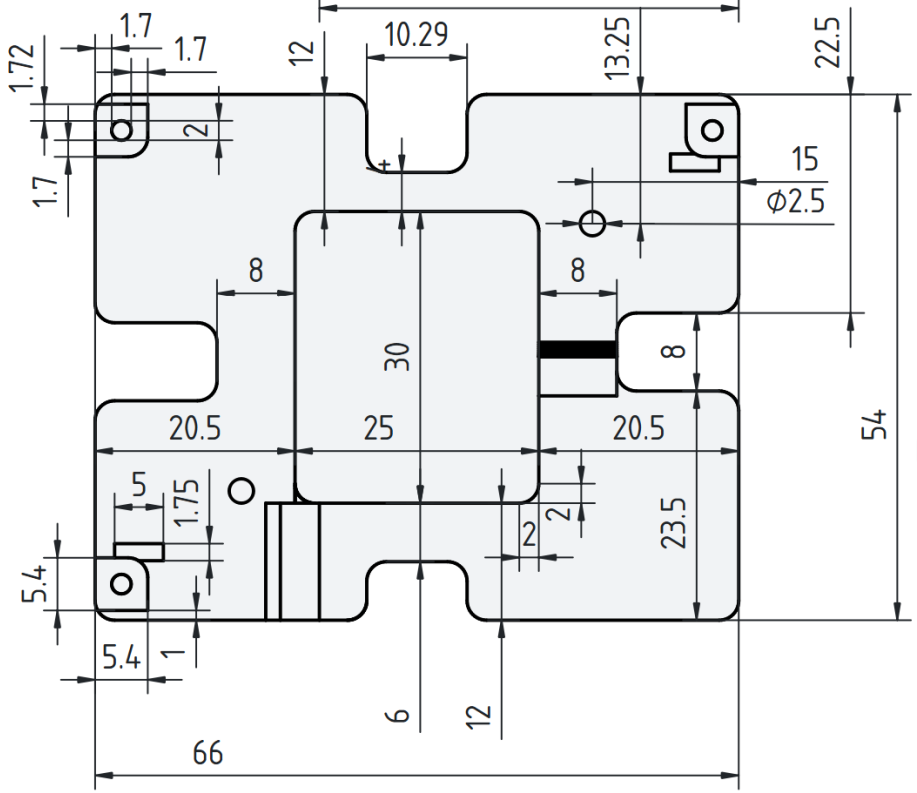


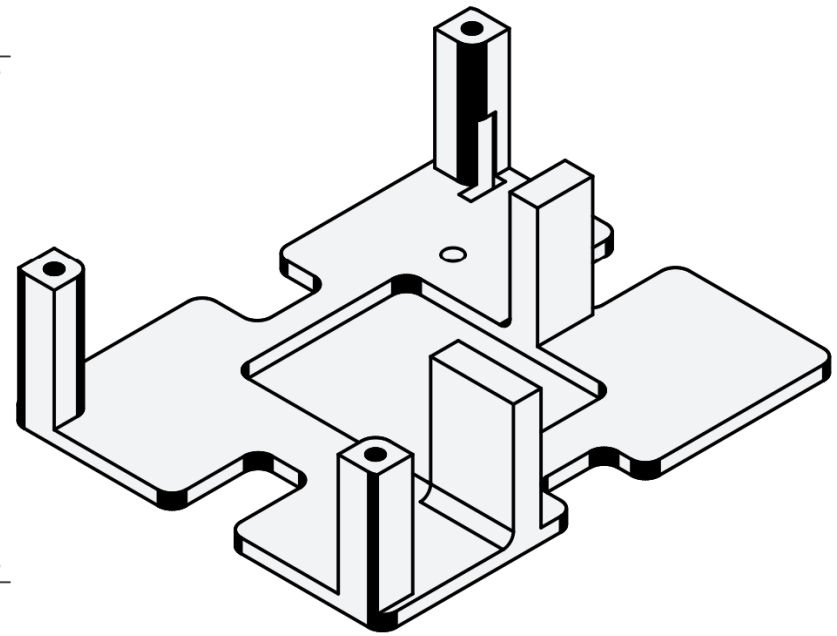
https://a.co/d/0ceDxgMU

Appendix B:
Various Part Drawings

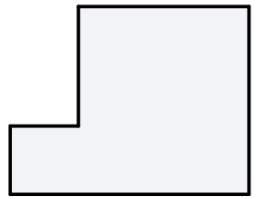


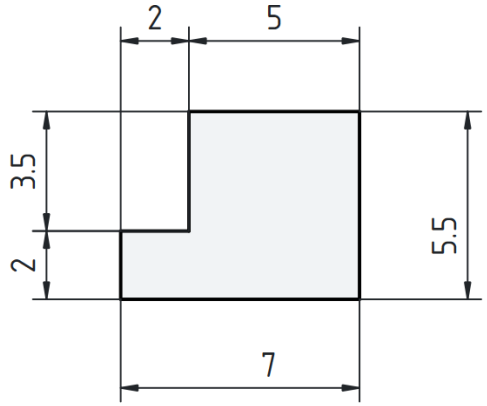


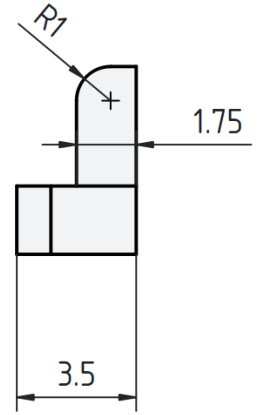


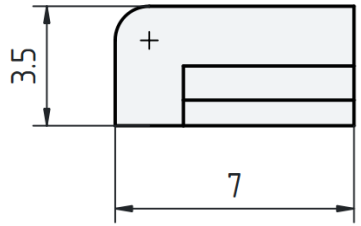


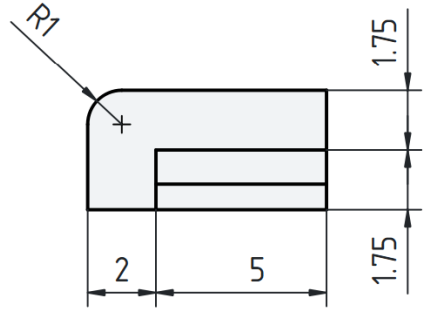
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Supplementary information: FreeCAD DRAWING		Size: A3	Sheet: X / Y	Scale: 2:1
		Part number: PN		
		Drawing number: DN		
		Date: 04/19/2026		Revision: REV A

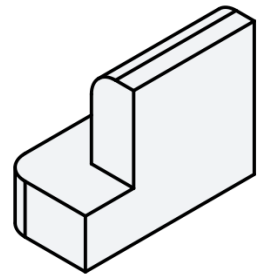


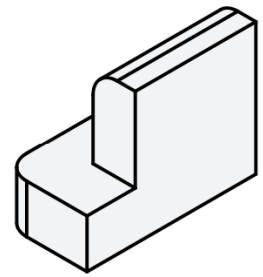




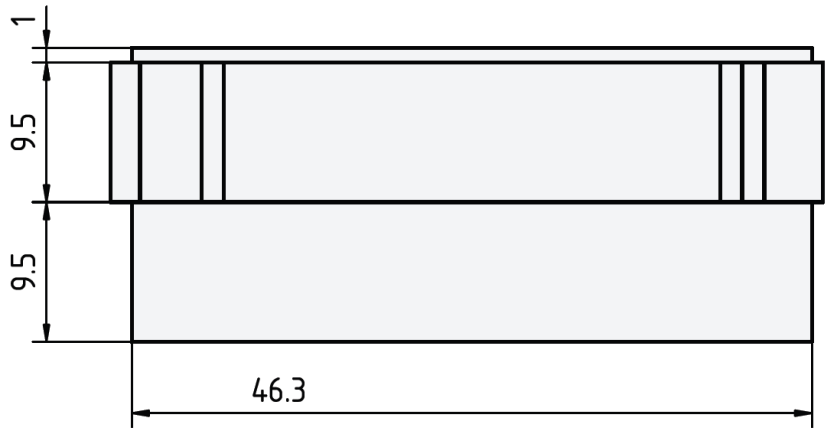


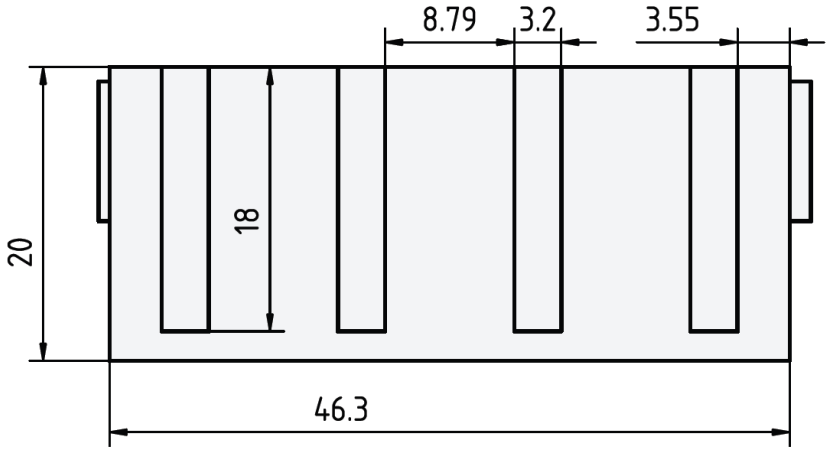


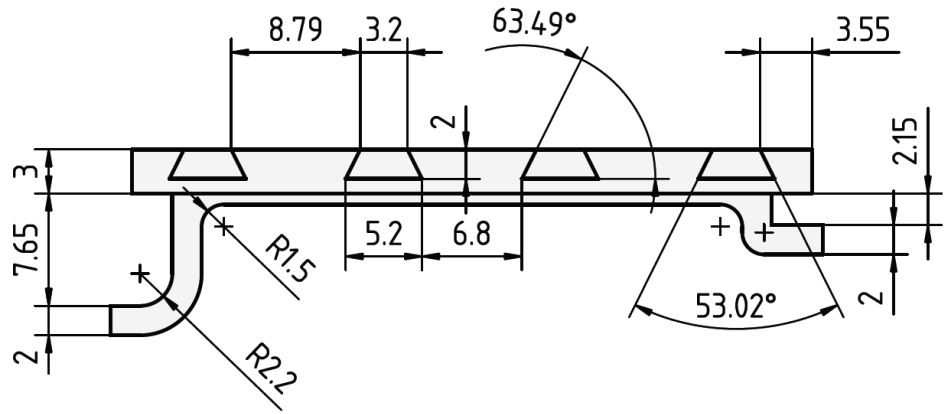


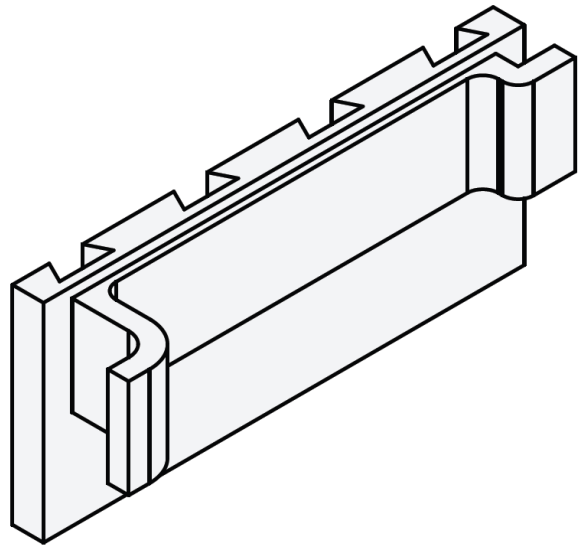


Created by: D. Warner	Title: Motor Locks (2x)			
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		Date: 04/19/2026		Revision: REV A

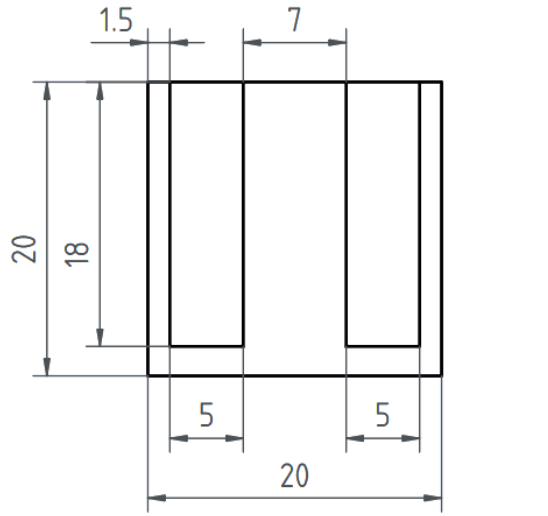


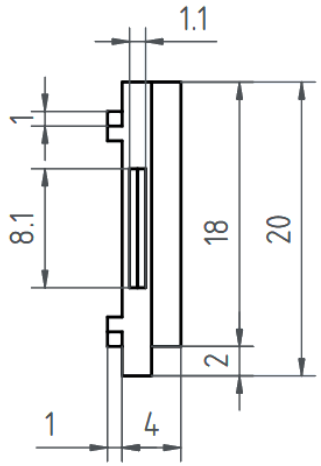


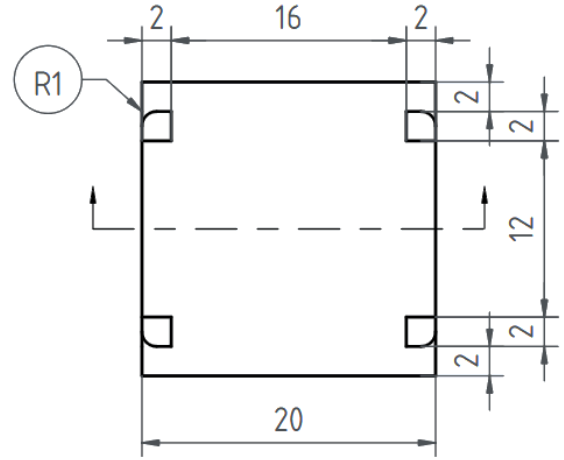


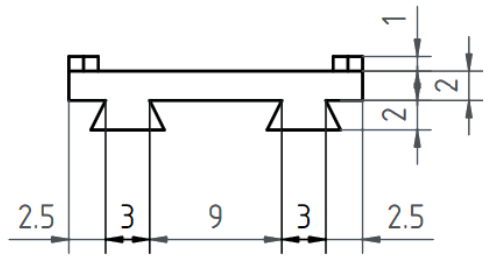


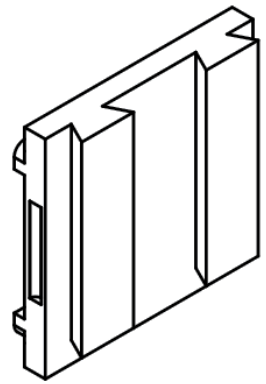
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Supplementary information: FreeCAD DRAWING	Size: A3	Sheet: X / Y	Scale: 3:1	
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	Date: 05/02/2026		Revision: REV A	

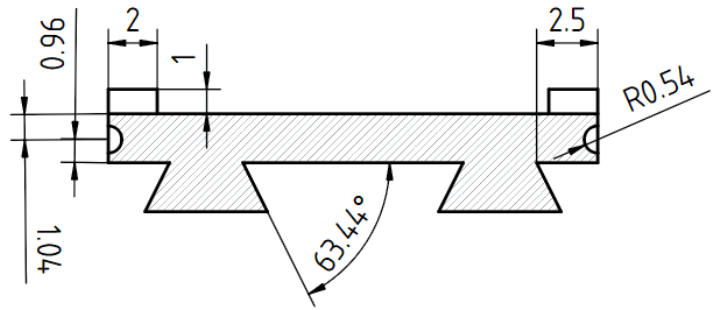




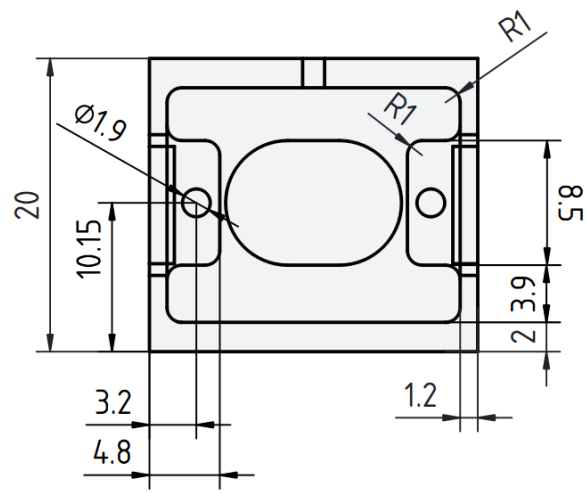


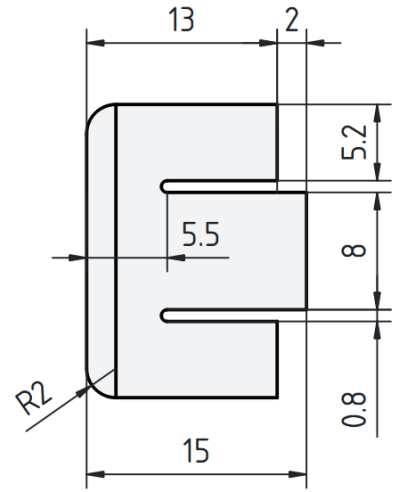


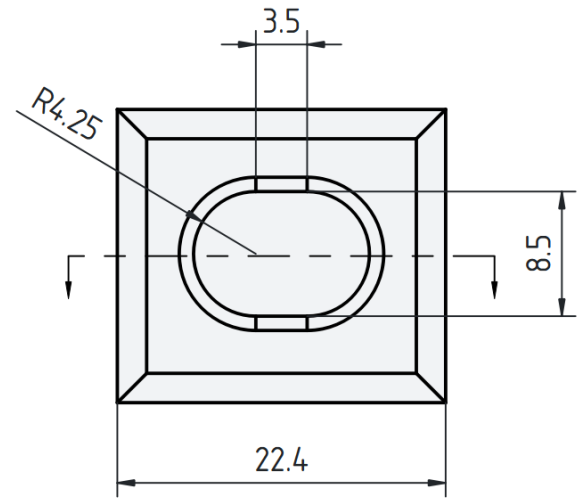


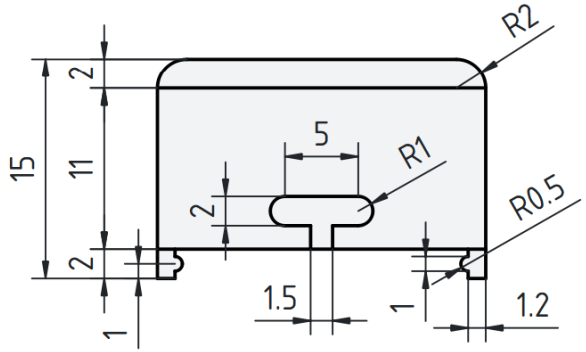


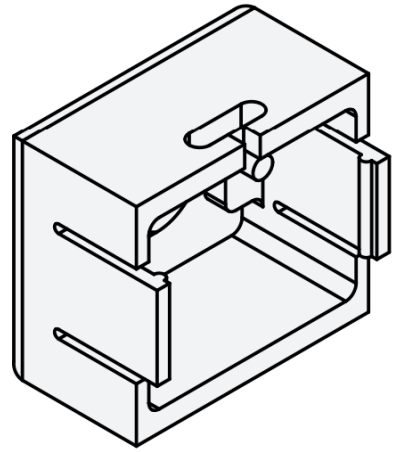
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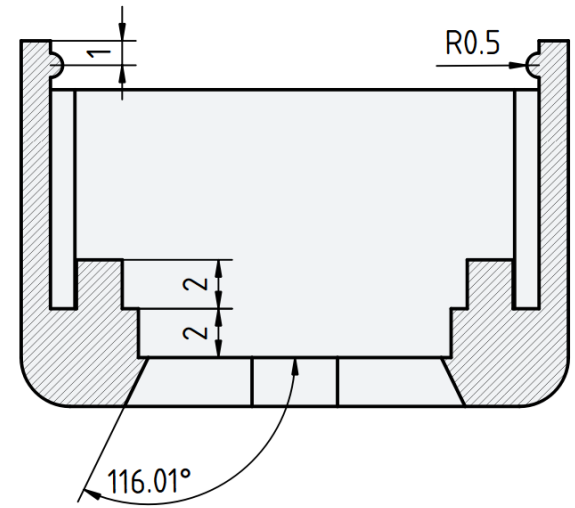












Created by: D. Warner	Title: VL531X Housing			
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